6

Strictly Positive Measures

We study in this chapter the condition of the existence of a strictly positive measure and its relation to the countable chain condition and to related conditions of calibre. A result of Kelley (Theorem 6.4) gives a condition (Kelley's property (**)) on the set of non-empty, open subsets of a compact, Hausdorff space which is necessary and sufficient for the existence of a strictly positive measure; we show by simple examples, given by Argyros, van Douwen, Koumoullis and Sapounakis, that not every completely regular, Hausdorff (non-compact) space satisfying Kelley's property has a strictly positive measure (Theorems 6.35 and 6.37).

The principal chain conditions satisfied by spaces with a strictly positive measure are the properties $K_{\alpha,n}$ for every cardinal α with $\mathrm{cf}(\alpha) > \omega$ and every natural number $n \geq 2$ (this is Theorem 6.15 and Corollary 6.17, of Argyros & Kalamidas). On the other hand there is a compact Hausdorff space with a strictly positive (regular, Borel) measure which, assuming the continuum hypothesis, does not have calibre ω^+. This example, noted by Erdős, is the Stone space of the Lebesgue measure algebra on [0, 1]. The generalization given here in Corollary 6.19 provides another example (supplementary to the space of Theorem 5.28) on the class of calibres of compact spaces.

We give three compact c.c.c. spaces which have no strictly positive measure and which, assuming (when necessary) the continuum hypothesis, have various differing calibre properties. These are as follows.

(a) A compact Hausdorff space with the countable chain condition, indeed with condition (*) defined below (a condition implied by Kelley's condition), but with no strictly positive measure. This example is due to Gaifman (Theorem 6.23). We show in addition that this space satisfies condition K_n for $2 \leq n < \omega$ and, assuming the continuum hypothesis, that it does not have calibre ω^+.

(b) For every natural number $n \geq 2$, a compact Hausdorff space with condition $(*)$, without a strictly positive measure, satisfying condition K_n, and, assuming the continuum hypothesis, not satisfying condition K_{n+1} (Argyros, Theorem 6.26).

(c) A compact Hausdorff space with calibre ω^+ and not satisfying the (ω, ω)-chain condition (hence with no strictly positive measure). This example, due to Galvin and Hajnal, graph-theoretic and combinatorial in nature, is given in Theorem 6.32.

Once it is recognized that a compact c.c.c. space need not have a strictly positive measure it is reasonable to ask if there is a cardinal number α with this property: For every compact c.c.c. space X there is a family $\{\mu_i : i < \alpha\}$ of regular, Borel measures on X such that for all $U \in \mathscr{T}^*(X)$ there is $i < \alpha$ satisfying $\mu_i(U) > 0$. An example of Argyros (essentially the example of 5.28 above) answers this question in the negative. A different example with similar properties can be obtained by modification of the space of Galvin and Hajnal defined in 6.32.

Characterization of spaces with a strictly positive measure

Several terms which appear in the following definition are defined and discussed in Appendix C.

Definition. A space X *has a strictly positive measure* if there are a pseudobase $\mathscr{B}$ for the topology of X and a probability measure μ, defined on the σ-field generated by $\mathscr{B}$, such that

$$\mu(B) > 0 \text{ for all non-empty } B \in \mathscr{B}.$$

Definition. A space X has *property* $(*)$ if the set $\mathscr{T}^*(X)$ of non-empty, open subsets of X can be written in the form

$$\mathscr{T}^*(X) = \bigcup_{n < \omega} \mathscr{T}_n$$

where, for $n < \omega$, no $n + 2$ elements of the family $\mathscr{T}_n$ are pairwise disjoint.

We note that a space with a strictly positive measure μ has property $(*)$. Indeed, if $\mathscr{B}$ is a pseudobase for the topology of X such that

$\mu(B) > 0$ for every non-empty $B \in \mathscr{B}$ and for $n < \omega$ we set

$\mathscr{T}_n = \{U \in \mathscr{T}^*(X)$: there is $B \in \mathscr{B}$ with $B \subset U$ and $\mu(B) \geq 1/(n+1)\}$,

then the relation

$$\mathscr{T}^*(X) = \bigcup_{n<\omega} \mathscr{T}_n$$

expresses $\mathscr{T}^*(X)$ in the required form.

Definition. Let X be a space.

(a) If $\mathscr{F} = \{U_1, \ldots, U_N\}$ is a non-empty, finite family of non-empty, open subsets of X, indexed (not necessarily faithfully) by $\{1, \ldots, N\}$, then cal $\mathscr{F}$ denotes the largest integer k such that there is $S \subset \{1, \ldots, N\}$ with $|S| = k$ such that $\cap_{i \in S} U_i \neq \varnothing$.

(b) If $\mathscr{S} \subset \mathscr{P}^*(X)$, then

$$\kappa(\mathscr{S}) = \inf\{(\text{cal } \mathscr{F})/N : 1 \leq N < \omega, \mathscr{F} = \{U_1, \ldots, U_N\} \subset \mathscr{S}\}.$$

(c) X has *property* (**) if the set $\mathscr{T}^*(X)$ of non-empty, open subsets of X can be written in the form

$$\mathscr{T}^*(X) = \bigcup_{n<\omega} \mathscr{T}_n$$

with $\kappa(\mathscr{T}_n) > 0$ for $n < \omega$.

Intuitively, property (*) is a finite version of a chain-type condition, while property (**) is a corresponding finite version of a calibre-type condition. The relation between the two, given in Lemma 6.2(b), is as expected.

6.1 *Lemma.* Let $(X, \mathscr{S}, \mu)$ be a probability measure space and $\mathscr{A}$ a family of (μ-measurable) subsets of X such that $\mu(A) > 0$ for $A \in \mathscr{A}$, and let $n < \omega$ and

$$\mathscr{U} \subset \{U \in \mathscr{P}^*(X): \text{there is } A \in \mathscr{A}, A \subset U, \mu(A) \geq 1/(n+1)\}.$$

Then $\kappa(\mathscr{U}) \geq 1/(n+1)$.

Proof. Let $\mathscr{F} = \{U_i : 1 \leq i \leq N\} \subset \mathscr{U}$ with $N < \omega$ and choose $A_i \in \mathscr{A}$ with $A_i \subset U_i$ and $\mu(A_i) \geq 1/(n+1)$ for $1 \leq i \leq N$.
For $S \subset \{1, 2, \ldots, N\}$ we set $S' = \{1, 2, \ldots, N\} \backslash S$ and we set

$$V_S = (\bigcap_{i \in S} A_i) \cap (\bigcap_{i \in S'} (X \backslash A_i)).$$

We note that if $S, T \subset \{1, 2, \ldots, N\}$ and $S \neq T$ then $V_S \cap V_T = \varnothing$.

We note further that

$$A_i = \cup\{V_S : i \in S\}$$

(indeed if $x \in A_i$ and $S(x) = \{j : x \in A_j\}$ then $i \in S(x)$ and $x \in V_{S(x)}$; and $V_S \subset A_i$ when $i \in S$). It follows that

$$\mu(A_i) = \sum\{\mu(V_S) : i \in S\} \quad \text{for } 1 \leq i \leq N,$$

and hence

$$N \cdot 1/(n+1) \leq \sum_{i=1}^{N} \mu(A_i) = \sum_{i=1}^{N} [\sum\{\mu(V_S) : i \in S\}];$$

for $V_S \neq \varnothing$ the summand $\mu(V_S)$ appears on the right-hand side with multiplicity $|S|$, and

$$|S| \leq \text{cal}\, \mathscr{F} \text{ when } V_S \neq \varnothing.$$

Thus we have

$$\begin{aligned} N \cdot 1/(n+1) &\leq (\text{cal}\, \mathscr{F}) \cdot \sum\{\mu(V_S) : S \subset \{1, 2, \ldots, N\}\} \\ &\leq (\text{cal}\, \mathscr{F}) \cdot 1 = \text{cal}\, \mathscr{F} \end{aligned}$$

and hence

$$(\text{cal}\, \mathscr{F})/N \geq 1/(n+1).$$

6.2 *Lemma*. Let X and Y be spaces.

(a) If X has a strictly positive measure then X has property (**).

(b) If X has property (**) then X has property (*).

(c) If $X \leq Y$ and Y has property (*), then X has property (*); in particular, X has property (*) if and only if its Gleason space ($G(X)$ has property (*).

(d) If $X \leq Y$ and Y has property (**) then X has property (**); in particular, X has property (**) if and only if its Gleason space $G(X)$ has property (**).

Proof. (a) Let μ be a strictly positive measure on X. Let $\mathscr{B}$ be a pseudobase for X such that $\mu(B) > 0$ for every non-empty $B \in \mathscr{B}$, and for $n < \omega$ set

$$\mathscr{T}_n = \{U \in \mathscr{T}^*(X) : \text{there is } B \in \mathscr{B} \text{ with } B \subset U \text{ and } \mu(B) \geq 1/(n+1)\}.$$

Then $\mathscr{T}^*(X) = \cup_{n<\omega} \mathscr{T}_n$, and from Lemma 6.1 (with $\mathscr{A}$ and $\mathscr{U}$ replaced by $\mathscr{B}$ and $\mathscr{T}_n$, respectively) it follows that $\kappa(\mathscr{T}_n) \geq 1/(n+1)$ for $n < \omega$.

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(b) Let $\mathscr{T}^*(X) = \cup_{n<\omega} \mathscr{U}_n$ with $\kappa(\mathscr{U}_n) > 0$ for $n < \omega$, and let $k(n)$ satisfy

$$\kappa(\mathscr{U}_n) > 1/(k(n)+2) \quad \text{and}$$
$$k(n) \neq k(n') \quad \text{for } n < n' < \omega.$$

We claim that no $k(n)+2$ elements of $\mathscr{U}_n$ are pairwise disjoint. Indeed if $\mathscr{F} \subset \mathscr{U}_n$ and $|\mathscr{F}| = k(n)+2$ and the elements of $\mathscr{F}$ are pairwise disjoint then cal $\mathscr{F} = 1$ and we have

$$1/(k(n)+2) < \kappa(\mathscr{U}_n) \leq \text{cal}\, \mathscr{F}/(k(n)+2) = 1/(k(n)+2),$$

a contradiction.

Finally we set

$$\mathscr{T}_{k(n)} = \mathscr{U}_n \quad \text{for } n < \omega, \quad \text{and}$$
$$\mathscr{T}_m = \varnothing \quad \text{if there is no } n < \omega \text{ with } m = k(n).$$

It is clear that

$$\mathscr{T}^*(X) = \bigcup_{n<\omega} \mathscr{T}_n, \quad \text{and}$$

no $n+2$ elements of the family $\mathscr{T}_n$ are pairwise disjoint.

Hence X has property (*).

The proof is complete.

(c) Since Y has property (*), the set $\mathscr{T}^*(Y)$ can be written in the form

$$\mathscr{T}^*(Y) = \bigcup_{n<\omega} \mathscr{T}_n$$

where, for $n < \omega$, no $n+2$ elements of the family $\mathscr{T}_n$ are pairwise disjoint. Since $X \leq Y$ there is $\varphi : \mathscr{T}^*(X) \to \mathscr{T}^*(Y)$ such that if $N < \omega$ and $\{U_k : k < N\} \subset \mathscr{T}^*(X)$ and $\cap_{k<N} U_k = \varnothing$, then $\cap_{k<N} \varphi(U_k) = \varnothing$.

We set

$$\mathscr{S}_n = \{U \in \mathscr{T}^*(X) : \varphi(U) \in \mathscr{T}_n\} \quad \text{for } n < \omega,$$

we note that $\mathscr{T}^*(X) = \cup_{n<\omega} \mathscr{S}_n$, and we note further that if $\mathscr{W}$ is a set of pairwise disjoint elements of $\mathscr{S}_n$ with $|\mathscr{W}| = n+2$, then $\{\varphi(U) : U \in \mathscr{W}\}$ is a set of pairwise disjoint elements of $\mathscr{T}_n$ with

$$|\{\varphi(U) : U \in \mathscr{W}\}| = n+2.$$

The second assertion of (c) now follows from the fact that $X \equiv G(X)$ (Theorem 2.20).

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(d) Since Y has property (**) the set $\mathscr{T}^*(Y)$ can be written in the form

$$\mathscr{T}^*(Y) = \bigcup_{n<\omega} \mathscr{T}_n$$

with $\kappa(\mathscr{T}_n) > 0$ for $n < \omega$. Since $X \leq Y$ there is $\varphi: \mathscr{T}^*(X) \to \mathscr{T}^*(Y)$ such that

if $N < \omega$ and $\{U_k : k < N\} \subset \mathscr{T}^*(X)$ and $\cap_{k<N} U_k = \varnothing$, then $\cap_{k<N} \varphi(U_k) = \varnothing$.

We set

$$\mathscr{S}_n = \{U \in \mathscr{T}^*(X) : \varphi(U) \in \mathscr{T}_n\} \quad \text{for } n < \omega$$

and we note that $\mathscr{T}^*(X) = \cup_{n<\omega} \mathscr{S}_n$. Now for $N < \omega$ and

$$\mathscr{F} = \{U_1, \dots, U_N\} \subset \mathscr{S}_n$$

we set

$$\mathscr{G} = \{\varphi(U_1), \dots, \varphi(U_N)\}$$

and we note from the characteristic property of the function φ that

$$\operatorname{cal} \mathscr{G} \leq \operatorname{cal} \mathscr{F}$$

and hence

$$(\operatorname{cal} \mathscr{G})/N \leq (\operatorname{cal} \mathscr{F})/N;$$

since $\mathscr{G} \subset \mathscr{T}_n$ we have

$$\kappa(\mathscr{S}_n) \geq \kappa(\mathscr{T}_n) > 0 \quad \text{for } n < \omega.$$

The second assertion of (d) now follows from the fact that $X \equiv G(X)$ (Theorem 2.20).

6.3 *Lemma.* Let X be a compact, Hausdorff space and let $\mathscr{S}$ be a non-empty family of non-empty, open subsets of X. Then there is a regular, Borel probability measure μ on X such that

$$\inf\{\mu(\operatorname{cl}_X U) : U \in \mathscr{S}\} \geq \kappa(\mathscr{S}).$$

Proof. We denote as usual by $C(X)$ the set of real-valued, continuous functions on X with norm

$$\| f \| = \sup\{|f(x)| : x \in X\} \quad \text{for } f \in C(X);$$

and we denote by F the set of elements f of $C(X)$ such that there are

$$n < \omega \quad \text{with } n \geq 1,$$
$$\{U_1, \dots, U_n\} \subset \mathscr{S},$$
$$\{t_1, \dots, t_n\} \subset [0, 1] \quad \text{with } \sum_{i=1}^{n} t_i = 1,$$
$$\{f_1, \dots, f_n\} \subset C(X) \quad \text{with } 0 \leq f_i(x) \leq 1$$

for $x \in X$, $1 \leq i \leq n$ and with $f_i | U_i \equiv 1$,

such that

$$f = t_1 f_1 + \dots + t_n f_n.$$

We claim first that if $f \in F$ then $\| f \| \geq \kappa(\mathscr{S})$.
Let $f \in F$ with

$$f = t_1 f_1 + \dots + t_n f_n$$

as above, let $\varepsilon > 0$, let $\{p_i : 1 \leq i \leq n\}$ be a set of positive integers with

$$\sum_{i=1}^{n} p_i = N, \quad \text{and}$$
$$\left\| \sum_{i=1}^{n} p_i f_i / N \right\| < \| f \| + \varepsilon,$$

and let $\mathscr{F}$ be the family, indexed by N, whose elements are the sets $U_i (1 \leq i \leq n)$ repeated with multiplicity p_i (i.e.,

$$\mathscr{F} = \{\underbrace{U_1, \dots, U_1}_{p_1 \text{ times}}, \dots, \underbrace{U_n, \dots, U_n}_{p_n \text{ times}}\}).$$

Since $\kappa(\mathscr{S}) \leq (\text{cal } \mathscr{F})/N$ and cal $\mathscr{F} \leq \| \sum_{i=1}^{n} p_i f_i \|$, we have

$$\kappa(\mathscr{S}) \leq \left\| \sum_{i=1}^{n} p_i f_i \right\| \Big/ N < N(\| f \| + \varepsilon)/N = \| f \| + \varepsilon;$$

it follows that $\kappa(\mathscr{S}) \leq \| f \|$.

The proof of the claim is complete.

Consider now the cone K in $C(X)$ defined by

$$K = \{\alpha(f + g) + \beta h : \alpha, \beta \geq 0, f \in F, g, h \in C(X), \| g \| < \kappa(\mathscr{S}), h \geq 0\}.$$

It is clear from the claim just proved that for $k \in K$ there is $x \in X$ such that $k(x) \geq 0$; hence K does not contain (the function with constant value equal to) -1 and there is by the Hahn–Banach theorem (cf. Section C. 11 of the Appendix) a bounded linear functional Λ on

$C(X)$ such that

$$-1 = \Lambda(-1) < \Lambda(k) \quad \text{for } k \in K.$$

From the fact that K is closed under multiplication by positive reals it then follows that

$$\Lambda(k) \geq 0 \quad \text{for } k \in K.$$

Now for $0 < \varepsilon < \kappa(\mathscr{S})$ we have

$$\|\varepsilon - \kappa(\mathscr{S})\| = \|\kappa(\mathscr{S}) - \varepsilon\| < \kappa(\mathscr{S})$$

and hence

$$f + \varepsilon - \kappa(\mathscr{S}) \in K \quad \text{for } f \in F,$$

so that

$$\Lambda(f) + \Lambda(\varepsilon - \kappa(\mathscr{S})) = \Lambda(f + \varepsilon - \kappa(\mathscr{S})) \geq 0 \quad \text{for } f \in F$$

and hence

$$\Lambda(f) \geq \Lambda(\kappa(\mathscr{S}) - \varepsilon) = \kappa(\mathscr{S}) - \varepsilon \quad \text{for } f \in F;$$

if follows that

$$\Lambda(f) \geq \kappa(\mathscr{S}) \quad \text{for } f \in F.$$

There is by the Riesz representation theorem (cf. Section C. 10 of the Appendix) a regular, Borel probability measure μ on X such that

$$\Lambda(f) = \int_X f \, d\mu \quad \text{for } f \in C(X).$$

We have

$$\mu(\mathrm{cl}_X U) = \inf \{\Lambda(f) : f \in C(X), f \geq 0, f|U \equiv 1\}$$

and hence

$$\mu(\mathrm{cl}_X U) \geq \kappa(\mathscr{S}) \quad \text{for } U \in \mathscr{S},$$

as required.

As a result we have the following theorem (in particular, the rather surprising implication (a) $\Rightarrow$ (b) below).

6.4 *Theorem.* Let X be a compact Hausdorff space. The following are equivalent:

(a) X has a strictly positive measure;

(b) X has a regular Borel strictly positive measure;

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(c) X satisfies condition (**); and

(d) the Gleason space $G(X)$ of X has a strictly positive measure.

Proof. (a)$\Rightarrow$(c) This is Lemma 6.2(a).

(c)$\Rightarrow$(b) Let $\mathscr{T}^*(X) = \cup_{n<\omega}\mathscr{T}_n$ with $\kappa(\mathscr{T}_n) > 0$ for $n < \omega$. There is by Lemma 6.2 a regular, Borel probability measure μ_n on X such that

$$\inf\{\mu_n(\mathrm{cl}\ U): U \in \mathscr{T}_n\} \geq \kappa(\mathscr{T}_n) > 0 \quad \text{for } n < \omega.$$

We set

$$\mu = \sum_{n<\omega} \mu_n/2^n;$$

then μ is a regular, Borel probability measure on X, and it is clear that μ is strictly positive on X.

(b)$\Rightarrow$(a) This is trivial.

(a) $\Leftrightarrow$(d) We have noted in Lemma 6.2(d) that X satisfies condition (**) if and only if $G(X)$ satisfies condition (**). The equivalence (a)$\Leftrightarrow$(d) then follows from the equivalence (a)$\Leftrightarrow$(c) of this theorem, already proved.

We give two consequences of Theorem 6.4.

6.5 *Corollary.* If a space X has a strictly positive measure, then its Gleason space $G(X)$ has a strictly positive measure.

Proof. It follows from Lemma 6.2 that X (and hence $G(X)$) have property (**). It then follows from Theorem 6.4((c)$\Rightarrow$(a)) (with X replaced by the compact Hausdorff space $G(X)$) that $G(X)$ has a strictly positive measure.

6.6 *Corollary.* For a completely regular, Hausdorff space X, the following statements are equivalent.

(a) The Gleason space $G(X)$ has a strictly positive measure;

(b) the Stone–Čech compactification βX has a strictly positive measure;

(c) some Hausdorff compactification of X has a strictly positive measure;

(d) every Hausdorff compactification of X has a strictly positive measure.

Proof. Let K be either the Gleason space of X or a Hausdorff

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compactification of X. From either Theorem 2.20 or Theorem 2.16(b) we have $K \equiv X$, so from Lemma 6.2(d) it follows that X has property (**) if and only if K has property (**). Then from the equivalence (a)$\Leftrightarrow$(c) of Theorem 6.4 (applied to K in place of X) it follows that each of the conditions (a), (b), (c) and (d) is equivalent to the condition that X has property (**).

We note that the converse of Corollary 6.5 for compact, Hausdorff spaces is given by Theorem 6.4; and the condition that a (completely regular, Hausdorff) space X has a strictly positive measure implies the various (equivalent) conditions of Corollary 6.6. We show in 6.35–6.37 below, however, that there are completely regular, Hausdorff spaces which satisfy the conditions of Corollary 6.6 and which have no strictly positive measure.

Spaces with a strictly positive measure

6.7 *Theorem.* Let $\{X_i : i \in I\}$ be a set of spaces, each with a strictly positive measure, and let $X = \prod_{i\in I} X_i$. Then X has a strictly positive measure.

Proof. For $i \in I$ there are a pseudobase $\mathscr{B}_i$ for the topology of X_i, and a probability measure μ_i defined on the σ-algebra $\mathscr{S}_i$ generated by $\mathscr{B}_i$, such that $\mu_i(B) > 0$ for all non-empty $B \in \mathscr{B}_i$. Let $\mathscr{B}$ be the set of finite intersections of elements of the family

$$\{\pi_i^{-1}(B) : B \in \mathscr{B}_i, i \in I\}$$

and let $\mathscr{S}$ be the σ-algebra on X generated by $\mathscr{B}$; then $\mathscr{B}$ is a pseudobase for the topology of X, and $\mathscr{S}$ is equal to the σ-algebra generated by

$$\{\pi_i^{-1}(S) : S \in \mathscr{S}_i, i \in I\}.$$

It is now clear from Theorem C.2 of the Appendix that the (product) measure μ defined on $\mathscr{S}$ is a strictly positive measure for X.

6.8 *Corollary.* Every product of separable spaces has a strictly positive measure.

Proof. It is enough to show that every separable space has a strictly positive measure. Let $\{x_n : n < \omega\}$ be dense in the space X and

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define

$$\mu : \mathscr{P}(X) \to [0, 1]$$

by $\mu(A) = \sum\{1/2^n : n < \omega, x_n \in A\}$. Then μ (or more accurately: its restriction to the σ-algebra of Borel sets of X) is a strictly positive Borel measure on X.

The following example shows that not every compact Hausdorff space with a strictly positive measure is equivalent (in the sense of the relation $\equiv$ of Chapter 2) to a product of a set of separable spaces.

The measure algebra $\mathscr{B} = \mathscr{M}_\lambda/\mathscr{N}_\lambda$ of a measure space $(X, \mathscr{S}, \lambda)$ is defined in (the statement of) Theorem C.6 of the Appendix.

6.9 *Theorem.* There is an extremally disconnected, compact, Hausdorff space Ω such that

Ω has a strictly positive regular Borel measure,

Ω is not separable,

$w(\Omega) = 2^\omega$, and

Ω is not homeomorphic to the Gleason space of a product of separable Hausdorff spaces.

Proof. Let λ denote (completed) Lebesgue measure on the interval $[0, 1]$ and let Ω be the Stone space of the measure algebra of $([0, 1], \mathscr{M}, \lambda)$ (with $\mathscr{M}$ the σ-algebra of Lebesgue measurable subsets of $[0, 1]$). It follows from Theorem C.6 of the Appendix that Ω is an extremally disconnected, compact Hausdorff space with a strictly positive regular Borel measure and (since λ is atomless) that Ω is not separable.

For $A \in \mathscr{M}$ and $n < \omega$ there are (since λ is regular) closed $K_n \subset A$ and open $U_n \supset A$ such that $\lambda(U_n \backslash K_n) < 1/n$; since the Borel set S defined by

$$S = (\bigcap_{n<\omega} U_n) \backslash (\bigcup_{n<\omega} K_n)$$

differs from A by a λ-null set, the sets A and S determine the same element of the measure algebra. It follows that the quotient Boolean algebra $\mathscr{A} = \mathscr{M}/\mathscr{N}$ satisfies $|\mathscr{A}| = 2^\omega$, and from Theorem B.14 of the Appendix we have

$$w(\Omega) = w(S(\mathscr{A})) = |\mathscr{A}| = 2^\omega.$$

Suppose now that Ω is homeomorphic to the Gleason space of a product $X = \prod_{i \in I} X_i$ with X_i a separable Hausdorff space. We assume without loss of generality that $|X_i|>1$ for $i\in I$. It then follows that for $i\in I$ there is a regular-open subset $U_i\subset X_i$ with $U_i\neq X_i$, and the function $i\to\pi_i^{-1}(U_i)$ is one-to-one from I into the set $\mathscr{R}(X_I)$ of regular-open subsets of X_I. Now from

$$|I|\leq|\mathscr{R}(X_I)|=w(S(\mathscr{R}(X_I)))=w(G(X_I))=w(\Omega)=2^\omega$$

we have $|I|\leq 2^\omega$ and hence X_I is separable (cf. Theorem B.2). It follows from Theorem B.18 that $G(X_I)$ and hence Ω are separable, contrary to what was proved above.

The proof is complete.

6.10 *Remark.* Let S be the Gleason space of the product $\Omega\times X$, where Ω is the space of Theorem 6.9 and X is the Gleason space of $[0,1]$. It can be shown that S has a strictly positive regular Borel measure, that S does not have a strictly positive normal Borel measure (cf. Theorem C.7 of the Appendix), that S is not separable and that $w(S)=2^\omega$.

6.11 *Problem* (H. Rosenthal). Is there an extremally disconnected, compact Hausdorff space X with $w(X)\leq 2^\omega$ such that X has a strictly positive measure but $C(X)$ is not isomorphic as a Banach space to $C(\Omega)$ for any space Ω with a strictly positive normal measure? We note in this connection that Argyros & Haydon (1981) have found such an example if the requirement $w(X)\leq 2^\omega$ is relaxed to $w(X)\leq(2^\omega)^+$. Indeed, in this case we may take for X the Gleason space of $\{0,1\}^{((2^\omega)^+)}$.

The properties $\mathrm{K}_{\alpha,n}$

We recall from Chapter 2 that for $\alpha\geq\omega$ and $2\leq n<\omega$, a space is said to have property $\mathrm{K}_{\alpha,n}$ if it has calibre (α,α,n). (Thus a space X has property $\mathrm{K}_{\alpha,n}$ if and only if for every set $\{U_\xi:\xi<\alpha\}$ of non-empty, open subsets of X there is $A\subset\alpha$ with $|A|=\alpha$ such that if $B\subset A$ and $|B|=n$ then $\cap_{\xi\in B}U_\xi\neq\varnothing$.)

A space has property K_n if it has property $\mathrm{K}_{\omega^+,n}$. Property K_2 is property (K).

We show in Corollary 6.17 below that a space with a strictly positive measure has property $\mathrm{K}_{\alpha,n}$ for all $\alpha\geq\omega$ with $\mathrm{cf}(\alpha)>\omega$ and for all n with $2\leq n<\omega$.

6.12 *Lemma.* Let $1 \leq n < \omega$, $1 \leq p < \omega$ and $1 \leq N < \omega$ with $p \leq 2^N$, and let $\{\theta_i : 1 \leq i \leq p\}$ be a set of positive real numbers. Then

(i) $(2p)^{n-1}\left(\sum_{i=1}^{p} \theta_i^n\right) > \left(\sum_{i=1}^{p} \theta_i\right)^n$, and

(ii) $\sum_{i=1}^{p} (1/2^N)\theta_i^n > (1/2^{n-1})\left(\sum_{i=1}^{p} (1/2^N)\theta_i\right)^n$.

Proof. (i) We proceed by induction on n. The statement is true for $n = 1$. We assume the statement true for $n = k < \omega$. We note for $i, j \in \{1, 2, \ldots, p\}$ that

$$\begin{aligned} \theta_i^{k+1} + \theta_j^{k+1} &\geq \theta_i^{k+1} \geq \theta_i\theta_j^k \quad \text{if } \theta_i \geq \theta_j \\ &\geq \theta_j^{k+1} \geq \theta_i\theta_j^k \quad \text{if } \theta_j \geq \theta_i \end{aligned}$$

and hence $\theta_i^{k+1} + \theta_j^{k+1} \geq \theta_i\theta_j^k$. It follows that

$$\begin{aligned} p\theta_i^{k+1} + \sum_{\substack{j=1 \\ j \neq i}}^{p} \theta_j^{k+1} &= \theta_i^{k+1} + \sum_{\substack{j=1 \\ j \neq i}}^{p} (\theta_i^{k+1} + \theta_j^{k+1}) \\ &\geq \theta_i^{k+1} + \sum_{\substack{j=1 \\ j \neq i}}^{p} \theta_i\theta_j^k = \theta_i\left[\sum_{j=1}^{p} \theta_j^k\right] \end{aligned}$$

and hence

$$\begin{aligned} (2p)^k\left(\sum_{i=1}^{p} \theta_i^{k+1}\right) &> (2p)^{k-1}(2p-1)\left(\sum_{i=1}^{p} \theta_i^{k+1}\right) \\ &= (2p)^{k-1}\left(\sum_{i=1}^{p}\left[p\theta_i^{k+1} + \sum_{\substack{j=1 \\ j \neq i}}^{p} \theta_j^{k+1}\right]\right) \\ &\geq (2p)^{k-1}\left(\sum_{i=1}^{p} \theta_i\left[\sum_{j=1}^{p} \theta_j^k\right]\right) \\ &= \left(\sum_{i=1}^{p} \theta_i\right)(2p)^{k-1}\left(\sum_{j=1}^{p} \theta_j^k\right) \\ &> \left(\sum_{i=1}^{p} \theta_i\right)\left(\sum_{i=1}^{p} \theta_i\right)^k = \left(\sum_{i=1}^{p} \theta_i\right)^{k+1}; \end{aligned}$$

the statement is proved for $n = k + 1$.

(ii) From (i) and the assumption $p \le 2^N$ we have

$$\begin{aligned}\sum_{i=1}^{p} (1/2^N)\theta_i^n &= (1/2^N)(1/(2p))^{n-1}(2p)^{n-1} \sum_{i=1}^{p} \theta_i^n \\ &> (1/2^N)(1/2^{n-1})(1/2^N)^{n-1}\left(\sum_{i=1}^{p} \theta_i\right)^n \\ &= (1/2^{n-1})\left(\sum_{i=1}^{p} (1/2^N)\theta_i\right)^n.\end{aligned}$$

6.13 *Lemma.* Let α be a cardinal with $\mathrm{cf}(\alpha) > \omega$, μ the Haar probability measure on $\{0, 1\}^I$, $\delta > 0$ and $\{U_\xi : \xi < \alpha\}$ a set of open-and-closed subsets of $\{0, 1\}^I$ such that $\mu(U_\xi) \ge \delta$ for $\xi < \alpha$. Then there is $A \subset \alpha$ with $|A| = \alpha$ such that $\mu(\cap_{\xi\in F} U_\xi) \ge \delta^n/4^{n-1}$ for all $F \subset A$ with $|F| \le n$.

Proof. Let $S(\xi)$ be a finite subset of I such that U_ξ depends on $S(\xi)$. We distinguish two cases.

Case 1. α is a regular cardinal. We show (for use in Case 2 below) that in fact $\mu(\cap_{\xi\in F} U_\xi) \ge \delta^n/2^{n-1}$ for $F \subset A$ with $|F| \le n$. By the Erdős–Rado theorem for quasi-disjoint families (Theorem 1.4) there are $C \subset \alpha$ with $|C| = \alpha$ and a set J such that

$$S(\xi) \cap S(\xi') = J \quad \text{for } \xi, \xi' \in C, \xi \ne \xi'.$$

We assume without loss of generality that $J \ne \varnothing$ and we set $|J| = N$. We set

$$\begin{aligned} H_\xi^x &= \{y \in \{0, 1\}^{S(\xi)\setminus J} : \langle x, y\rangle \in \pi_{S(\xi)}[U_\xi]\} \\ &\qquad \text{for } x \in \{0, 1\}^J, \xi \in C, \quad \text{and} \\ T_\xi &= \{x \in \{0, 1\}^J : H_\xi^x \ne \varnothing\} \quad \text{for } \xi \in C.\end{aligned}$$

We note that

$$\begin{aligned}\pi_{S(\xi)}[U_\xi] &= \cup\{\{x\} \times H_\xi^x : x \in \{0, 1\}^J\} \\ &= \cup\{\{x\} \times H_\xi^x : x \in T_\xi\} \quad \text{for } \xi \in C.\end{aligned}$$

It then follows, denoting by μ_ξ the Haar probability measure on $\{0, 1\}^{S(\xi)\setminus J}$, that

$$\text{(1)} \qquad \delta \le \mu(U_\xi) = (1/2^N)(\textstyle\sum\{\mu_\xi(H_\xi^x) : x \in T_\xi\}) \quad \text{for } \xi \in C.$$

Since α is an infinite cardinal there are $B \subset C$ with $|B| = \alpha$ and a set $T = \{x_i : 1 \le i \le p\} \subset \{0, 1\}^J$ such that

$$T_\xi = T \quad \text{for } \xi \in B;$$

we note that $|T| = p \le 2^N$. Finally since $\mathrm{cf}(\alpha) > \omega$ there are $A \subset B$ with $|A| = \alpha$ and a set $\{\theta_i : 1 \le i \le p\}$ of positive rational numbers such that

$$\text{(2)} \qquad \mu_\xi(H_\xi^{x_i}) = \theta_i \text{ for } 1 \le i \le p,\ \xi \in A.$$

Now let $F \subset A$ with $|F| = n$ and set $S = \cup_{\xi \in F} S(\xi)$. Since

$$\bigcap_{\xi \in F} U_\xi = \left(\bigcup_{i=1}^{p} \left(\{x_i\} \times \prod_{\xi \in F} H_\xi^{x_i} \right) \right) \times \{0, 1\}^{I \setminus S}$$

we have from (1), (2) and Lemma 6.10 that

$$\mu\left(\bigcap_{\xi \in F} U_\xi \right) = \sum_{i=1}^{p} (1/2^N) \left(\prod_{\xi \in F} \mu_\xi(H_\xi^{x_i}) \right) = \sum_{i=1}^{p} (1/2^N) \theta_i^n$$
$$\ge (1/2^{n-1}) \left(\sum_{i=1}^{p} (1/2^N) \theta_i \right)^n \ge \delta^n / 2^{n-1},$$

as required.

Case 2. α is a singular cardinal. There is by Theorem A.7 of the Appendix a set $\{\alpha_\sigma : \sigma < \mathrm{cf}(\alpha)\}$ of regular cardinals such that

$$\omega < \alpha_\sigma < \alpha_{\sigma'} < \alpha \quad \text{for } \sigma < \sigma' < \mathrm{cf}(\alpha), \text{ and}$$
$$\alpha = \sum_{\sigma < \mathrm{cf}(\alpha)} \alpha_\sigma.$$

By the extension to singular cardinals of the Erdős–Rado theorem (Theorem 1.9), there are Γ, J, $\{D_\sigma : \sigma \in \Gamma\}$ and $\{J_\sigma : \sigma \in \Gamma\}$ such that

$$\Gamma \subset \mathrm{cf}(\alpha) \quad \text{and } |\Gamma| = \mathrm{cf}(\alpha),$$
$$D_\sigma \subset \alpha \quad \text{and } |D_\sigma| = \alpha_\sigma \quad \text{for } \sigma \in \Gamma,$$
$$D_\sigma \cap D_{\sigma'} = \varnothing \quad \text{for } \sigma', \sigma \in \Gamma, \sigma \ne \sigma', \quad \text{and}$$
$$S(\xi) \cap S(\xi') = J_\sigma \quad \text{for } \xi, \xi' \in D_\sigma, \xi \ne \xi'$$
$$= J \quad \text{for } \xi \in D_\sigma, \xi' \in D_{\sigma'}, \sigma, \sigma' \in \Gamma, \sigma \ne \sigma'.$$

We assume without loss of generality that $J \ne \varnothing$ and we set $|J| = N$.

We set $D = \cup_{\sigma < \mathrm{cf}(\alpha)} D_\sigma$, as before we set

$$H_\xi^x = \{y \in \{0,1\}^{S(\xi)\setminus J} : \langle x, y\rangle \in \pi_{S(\xi)}[U_\xi]\}$$
$$\text{for } x \in \{0,1\}^J, \xi \in D, \quad \text{and}$$
$$T_\xi = \{x \in \{0,1\}^J : H_\xi^x \neq \emptyset\} \quad \text{for } \xi \in D,$$

and we note that

$$\pi_{S(\xi)}[U_\xi] = \cup\{\{x\} \times H_\xi^x : x \in T_\xi\} \quad \text{for } \xi \in D.$$

For $\sigma \in \Gamma$ there are $C_\sigma \subset D_\sigma$ with $|C_\sigma| = \alpha_\sigma$ and $T(\sigma) \subset \{0,1\}^J$ such that

$$T_\xi = T(\sigma) = \{x_i(\sigma) : 1 \leq i \leq p(\sigma)\} \quad \text{for } \xi \in C_\sigma,$$

and there are $B_\sigma \subset C_\sigma$ with $|B_\sigma| = \alpha_\sigma$ and set $\{\theta_i(\sigma) : 1 \leq i \leq p(\sigma)\}$ of positive rational numbers such that, denoting by μ_ξ the Haar probability measure on $\{0,1\}^{S(\xi)\setminus J}$, we have

$$\mu_\xi(H_\xi^{x_i(\sigma)}) = \theta_i(\sigma) \quad \text{for } 1 \leq i \leq p(\sigma), \xi \in B_\sigma.$$

Since $\mathrm{cf}(\alpha) > \omega$ there are $\Delta \subset \Gamma$ with $|\Delta| = \mathrm{cf}(\alpha)$ and a set $T = \{x_i : 1 \leq i \leq p\} \subset \{0,1\}^J$ and a set $\{\theta_i : 1 \leq i \leq p\}$ of positive rational numbers such that

$$p(\sigma) = p \quad \text{and } T(\sigma) = T \quad \text{for } \sigma \in \Delta, \quad \text{and}$$
$$\theta_i(\sigma) = \theta_i \quad \text{for } 1 \leq i \leq p, \quad \sigma \in \Delta.$$

Since

$$\mu_\xi(H_\xi^{x_i}) = \mu(H_\xi^{x_i} \times \{0,1\}^{(I\setminus S(\xi)) \cup J}) = \theta_i \quad \text{for } \xi \in B_\sigma$$

and α_σ is a regular, uncountable cardinal number, there is by Case 1 a set $A_\sigma \subset B_\sigma$ with $|A_\sigma| = \alpha_\sigma$ such that

$$(3) \qquad \mu(\bigcap_{\xi \in F} (H_\xi^{x_i} \times \{0,1\}^{(I\setminus S(\xi)) \cup J}))$$
$$\geq \theta_i^{|F|}/2^{|F|-1} \quad \text{for } F \subset A_\sigma, |F| \leq n.$$

We set $A = \cup_{\sigma \in \Delta} A_\sigma$ and we note $|A| = \alpha$.

Now let $F \subset A$ with $|F| = n$, and set $S = \cup_{\xi \in F} S(\xi)$,

$$\{\sigma \in \Delta : F \cap A_\sigma \neq \emptyset\} = \{\sigma_j : 1 \leq j \leq m\} \text{ with } 1 \leq m \leq n, \text{ and}$$
$$F_j = F \cap A_{\sigma_j} \quad \text{for } 1 \leq j \leq m.$$

Since

$$\bigcap_{\xi \in F} U_\xi = \bigcap_{j=1}^{m} (\bigcap_{\xi \in F_j} U_\xi)$$
$$= \bigcup_{i=1}^{p} (\{x_i\} \times \prod_{j=1}^{m} (\bigcap_{\xi \in F_j} H_\xi^{x_i} \times \{0,1\}^{(I\setminus S(\xi)) \cup J}))$$

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we have from (3) and Lemma 6.12 that

$$\begin{aligned}\mu\Big(\bigcap_{\xi\in F} U_\xi\Big) &= \sum_{i=1}^{p} (1/2^N) \prod_{j=1}^{m} \theta_i^{|F_j|}/2^{|F_j|-1} \\ &= \sum_{i=1}^{p} (1/2^N)\theta_i^n/2^{n-m} \geq 1/2^{n-1} \sum_{i=1}^{p} (1/2^N)\theta_i^n \\ &\geq (1/2^{n-1})(1/2^{n-1})\Big(\sum_{i=1}^{p} (1/2^N)\theta_i\Big)^n \\ &\geq \delta^n/4^{n-1},\end{aligned}$$

as required.

6.14 *Corollary.* Let α be a cardinal with $\mathrm{cf}(\alpha) > \omega$, μ the Haar probability measure on $\{0, 1\}^I$, and $\{B_\xi : \xi < \alpha\}$ a set of Borel subsets of $\{0, 1\}^I$ such that $\mu(B_\xi) > 0$ for $\xi < \alpha$. Then for $1 \leq n < \omega$ there are $A_n \subset \alpha$ with $|A_n| = \alpha$ and $\delta_n > 0$ such that $\mu(\bigcap_{\xi\in F} B_\xi) \geq \delta_n$ for all $F \subset A_n$ with $|F| \leq n$.

Proof. We assume without loss of generality that there is $\delta > 0$ such that $\mu(B_\xi) > \delta$ for $\xi < \alpha$; and we set $\delta_n = (\delta/8)^n/2$.

Since the Borel measure μ is regular (cf. Appendix C), for $\xi < \alpha$ there is an open-and-closed subset U_ξ of $\{0, 1\}^I$ such that

$$\mu(U_\xi \Delta B_\xi) < (\delta/8)^n/(2n) \leq (\delta/8)^n/2 < \delta/2$$

and hence

$$\mu(U_\xi) \geq \mu(B_\xi) - \mu(U_\xi \Delta B_\xi) > \delta - \delta/2 = \delta/2.$$

It follows from Lemma 6.13 that there is $A_n \subset A$ with $|A_n| = \alpha$ such that if $F \subset A_n$ with $|F| = n$ then

$$\mu\Big(\bigcap_{\xi\in F} U_\xi\Big) \geq (\delta/2)^n/4^{n-1};$$

from

$$\bigcap_{\xi\in F} U_\xi \subset \Big(\bigcap_{\xi\in F} B_\xi\Big) \cup \Big(\bigcup_{\xi\in F} (U_\xi \Delta B_\xi)\Big)$$

we then have

$$\begin{aligned}(\delta/8)^n < (\delta/2)^n/4^{n-1} \leq \mu\Big(\bigcap_{\xi\in F} U_\xi\Big) &\leq \mu\Big(\bigcap_{\xi\in F} B_\xi\Big) + \sum_{\xi\in F} \mu(U_\xi \Delta B_\xi) \\ &\leq \mu\Big(\bigcap_{\xi\in F} B_\xi\Big) + (\delta/8)^n/2\end{aligned}$$

and hence

$$\mu\Big(\bigcap_{\xi\in F} B_\xi\Big) \geq (\delta/8)^n/2 = \delta_n,$$

as required.

6.15 *Theorem* (Argyros and Kalamidas). Let α be a cardinal with $\mathrm{cf}(\alpha) > \omega$, $(X, \mathscr{S}, \mu)$ a probability measure space and $\{S_\xi : \xi < \alpha\} \subset \mathscr{S}$ with $\mu(S_\xi) > 0$ for $\xi < \alpha$. Then for $1 \leq n < \omega$ there are $A_n \subset A$ with $|A_n| = \alpha$ and $\delta_n > 0$ such that $\mu(\cap_{\xi \in F} S_\xi) \geq \delta_n$ for all $F \subset A_n$ with $|F| \leq n$.

Proof. Let $(\mathscr{B}, \lambda)$ be the measure algebra of $(X, \mathscr{S}, \mu)$(cf. Theorem C.6 of the Appendix) and set $\Omega = S(\mathscr{B})$. It is by C.6 enough to show that if $\{V_\xi : \xi < \alpha\} \subset \mathscr{B}$ with $\lambda(V_\xi) > 0$ for $\xi < \alpha$ then for $1 \leq n \leq \omega$ there are $A_n \subset \alpha$ with $|A_n| = \alpha$ and $\delta_n > 0$ such that $\lambda(\cap_{\xi \in F} V_\xi) \geq \delta_n$ for all $F \subset A_n$ with $|F| \leq n$.

Now let $\gamma, \delta, \{p_i : i < \gamma\}, \{\Omega_m : m < \delta\}, \{\alpha_m : m < \delta\}$ and $\{\mu_m : m < \delta\}$ be as in the statement of Maharam's structure theorem (C.8 of the Appendix). Since

$$(\bigcup_{n<\delta} \Omega_n) \cup \{p_i : i < \gamma\}$$

is dense in Ω and $\gamma + \delta \leq \omega < \mathrm{cf}(\alpha)$, either there are $A \subset \alpha$ with $|A| = \alpha$ and $i < \gamma$ such that $p_i \in \cap_{\xi \in A} V_\xi$, or there are $A \subset \alpha$ with $|A| = \alpha$ and $m < \delta$ such that $V_\xi \cap \Omega_m \neq \varnothing$ for $\xi \in A$. In the first case we set $\delta_n = \lambda(\{p_i\}) > 0$ and the proof is complete. In the second case it is enough to show that if $\{B_\xi : \xi \in A\}$ is a family of Borel subsets of $\{0, 1\}^{\alpha_m}$ with $\mu_m(B_\xi) > 0$ for $\xi \in A$, then for $1 \leq n < \omega$ there are $A_n \subset A$ with $|A_n| = \alpha$ and $\delta_n > 0$ such that $\mu_m(\cap_{\xi \in F} B_\xi) \geq \delta_n$ for all $F \subset A_n$ with $|F| \leq n$. This is the content of Corollary 6.14.

The proof is complete.

Remark. We note in passing that the proof of Theorem 6.15 does not require Maharam's structure theorem in case the (uncountable) cardinal number α is regular. Indeed for $n = 1$ the statement follows for arbitrary α from the assumption $\mathrm{cf}(\alpha) > \omega$, and then for regular $\alpha > \omega$ the proof is immediate (by induction) from the following lemma.

6.16 *Lemma.* Let α be an (infinite) regular cardinal, $(X, \mathscr{S}, \mu)$ a probability measure space, $2 \leq n < \omega$, $\{S_\xi : \xi \in A\} \subset \mathscr{S}$ with $|A| = \alpha$, $\delta > 0$ and k an integer such that $k\delta/2 > 1$. If

$$\mu(\bigcap_{\xi \in F} S_\xi) \geq \delta \quad \text{for all } F \subset A \text{ with } |F| < n$$

then there is $B \subset A$ with $|B| = \alpha$ such that

$$(1) \qquad \mu(\bigcap_{\xi \in F} S_\xi) \geq \delta/(4k) \quad \text{for all } F \subset B \text{ with } |F| \leq n.$$

Proof. Suppose the statement fails. We set $A_1 = A$ and we set

$$\mathscr{C}_1 = \{B : B \subset A_1, \{S_\xi : \xi \in B\} \text{ satisfies (1)}\}.$$

For $C \subset A_1$ with $|C| \leq n-1$ we have $C \in \mathscr{C}_1$ and hence $\mathscr{C}_1 \neq \varnothing$; further the set $\mathscr{C}_1$ partially ordered by inclusion is inductive. Hence there is a maximal element C_1 of $\mathscr{C}_1$. We have $|C_1| < \alpha$.

For $\xi \in A_1 \backslash C_1$ there is $F(\xi) \subset C_1$ with $|F(\xi)| < n$ such that

$$\mu(S_\xi \cap \bigcap_{\zeta \in F(\xi)} S_\zeta) < \delta/(4k),$$

and since α is regular and $|C_1| < \alpha$ there are $A_2 \subset A_1$ with $|A_2| = \alpha$ and a (fixed) set $\bar{F}(1) \subset C_1$ with $|\bar{F}(1)| < n$ such that, defining

$$T(1) = \bigcap_{\zeta \in \bar{F}(1)} S_\zeta,$$

we have

$$\mu(S_\xi \cap T(1)) < \delta/(4k) \quad \text{for } \xi \in A_2.$$

We repeat this argument with A_2 in place of A_1 and we find $C_2 \subset A_2$ with $|C_2| < \alpha$ and $A_3 \subset A_2$ with $|A_3| = \alpha$ and $\bar{F}(2) \subset C_2$ with $|\bar{F}(2)| < n$ such that, setting

$$T(2) = \bigcap_{\zeta \in \bar{F}(2)} S_\zeta,$$

we have

$$\mu(S_\xi \cap T(2)) < \delta/(4k) \quad \text{for } \xi \in A_3.$$

Repeating the argument we finally find $\{A_m : 1 \leq m \leq k+1\}$ with

$$A_{k+1} \subset A_k \subset \ldots \subset A_1 \quad \text{and}$$

$$|A_m| = \alpha,$$

and we find $\bar{F}(m) \subset C_m \subset A_m$ with $|\bar{F}(m)| < n$ such that, defining

$$T(m) = \bigcap_{\zeta \in \bar{F}(m)} S_\zeta,$$

we have

$$\mu(S_\xi \cap T(m)) < \delta/(4k) \quad \text{for } \xi \in A_{m+1}.$$

For $1 \le m < j \le k$ there is $\zeta \in \bar{F}(j) \subset A_j \subset T_m$ and hence

$$\mu(T(j) \cap T(m)) \le \mu(S_\xi \cap T(m)) < \delta/(4k);$$

it then follows, defining $U(1) = T(1)$ and $U(j) = T(j) \backslash \cup_{m=1}^{j-1} T(m)$ for $1 < j \le k$, that

$$\begin{aligned} &\mu(U(1)) \ge \delta \quad \text{and} \\ &\mu(U(j)) \ge \delta - (j-1)\delta/(4k) \\ &\qquad\qquad \ge \delta - (k-1)\delta/(4k) > \delta/2 \quad \text{for } 1 < j \le k. \end{aligned}$$

Since $U(m) \cap U(j) = \varnothing$ for $1 \le m < j \le k$ we have

$$\mu(X) \ge \sum_{j=1}^{k} \mu(U(j)) \ge k\delta/2 > 1,$$

a contradiction.

The proof is complete.

The next result, a consequence of Theorem 6.15, is the principal result of this section.

6.17 *Corollary.* Every space with property (**) (in particular: every space with a strictly positive measure) has property $K_{\alpha,n}$ for all cardinals α with $\mathrm{cf}(\alpha) > \omega$ and for all natural numbers $n \ge 2$.

Remark. Corollary 6.17 gives (strong) chain conditions satisfied by every space with a strictly positive measure. The weakest chain condition stronger than property $K_{\alpha,n}$ for all n is precalibre α. We show next that for $\mathrm{cf}(\beta) = \omega$ and assuming $\beta^+ = 2^\beta$, a space with a strictly positive measure need not have (pre-) calibre β; in particular, assuming the continuum hypothesis, there is a space with a strictly positive measure that does not have precalibre ω^+.

6.18 *Theorem.* Let β be a strong limit cardinal with $\mathrm{cf}(\beta) = \omega$, and assume that $2^\beta = \beta^+$. Then there is an extremally disconnected compact Hausdorff space Ω such that

Ω has a strictly positive regular Borel measure (and hence satisfies c.c.c.), and

Ω does not have calibre β^+.

Proof. Let Ω be the Stone space of the measure algebra of $(\{0, 1\}^\beta, \mathscr{S}, \lambda)$, where λ is the (complete) Haar probability measure on the compact Abelian group $\{0, 1\}^\beta$ and $\mathscr{S}$ is the σ-algebra of λ-measurable subsets of $\{0, 1\}^\beta$. It follows from Theorem C.6 of the Appendix that Ω is an extremally disconnected compact Hausdorff space with a strictly positive regular Borel measure.

We claim first that if $A \subset \{0, 1\}^\beta$ with $|A| \leq \beta$ then $A \in \mathscr{S}$ and $\lambda(A) = 0$. We note that if $|A| = \beta$ then since $\operatorname{cf}(\beta) = \omega$ there is a sequence $\{A_n : n < \omega\}$ such that $A = \cup_{n<\omega} A_n$ and $|A_n| < \beta$ for $n < \omega$; hence it is enough to prove the claim in the special case that $|A| < \beta$.

Let B be the smallest closed subgroup of $\{0, 1\}^\beta$ containing A. Then $B \in \mathscr{S}$, and since β is a strong limit cardinal we have

$$|B| \leq 2^{2^{|A|}} < \beta;$$

hence the number of cosets in the quotient $\{0, 1\}^\beta$ modulo B is certainly infinite, and since the probability measure λ is translation-invariant (cf. Theorem C.8), we have $\lambda(B) = 0$. Since λ is complete, from $A \subset B$ we have $A \in \mathscr{S}$ and $\lambda(A) = 0$, as required.

We show next that there is a family $\{K_\eta : \eta < \beta^+\}$ of subsets of $\{0, 1\}^\beta$ such that

K_η is compact for $\eta < \beta^+$,

$\lambda(K_\eta) > 0$ for $\eta < \beta^+$, and

if $A \subset \beta^+$ with $|A| = \beta^+$ then $\{K_\eta : \eta \in A\}$ does not have the finite intersection property.

Since $\beta^+ = 2^\beta$ there is a faithful indexing $\{x_\xi : \xi < \beta^+\}$ of $\{0, 1\}^\beta$ by β^+. We note that the set

$$\{x_\xi : \xi < \eta\}$$

has cardinality $|\eta| \leq \beta$ for all $\eta < \beta^+$, and hence from the claim proved above we have

$$\lambda(\{x_\xi : \xi < \eta\}) = 0 \quad \text{for } \eta < \beta^+.$$

Since the (complete) Haar measure λ is regular (cf. Theorem C.3), there is for $\eta < \beta^+$ a compact subset K_η of $\{x_\xi : \xi \geq \eta\}$ such that $\lambda(K_\eta) > 0$. Let $A \subset \beta^+$ with $|A| = \beta^+$. If $\{K_\eta : \eta \in A\}$ has the finite intersection property, then by compactness there is

$$x \in \bigcap_{\eta \in A} K_\eta.$$

Then $x = x_\xi$ for some $\xi < \beta^+$, and since $|A| = \beta^+$, there is $\eta \in A$ such that $\eta > \xi$; we have $x \notin K_\eta$, a contradiction. The proof that the family $\{K_\eta : \eta < \beta^+\}$ is as required is complete.

Now for $\eta < \beta^+$ let $[K_\eta]$ denote the $\mathcal{N}_\lambda$-equivalence class of K_η (where $\mathcal{N}_\lambda = \{A \in \mathcal{S} : \lambda(A) = 0\}$), so that $K_\eta \in [K_\eta] \in \mathcal{A} = \mathcal{M}_\lambda/\mathcal{N}_\lambda$, and let U_η denote the unique open-and-closed subset of Ω that corresponds to $[K_\eta]$ via Stone's duality.

We claim finally that if $A \subset \beta$ with $|A| = \beta^+$, then $\cap_{\eta \in A} U_\eta = \varnothing$. It is enough to prove that $\{U_\eta : \eta \in A\}$ does not have the finite intersection property, i.e., that $\{[K_\eta] : \eta \in A\}$ does not have the finite intersection property. This follows from the fact that $\{K_\eta : \eta \in A\}$ does not have the finite intersection property.

The proof is complete.

6.19 *Corollary.* Assume the generalized continuum hypothesis, and let α be a cardinal number such that

$$\mathrm{cf}(\alpha) = \beta^+ \quad \text{and } \mathrm{cf}(\beta) = \omega.$$

Then there is an extremally disconnected compact Hausdorff space Ω such that

Ω has a strictly positive regular Borel measure (and hence satisfies c.c.c.), and

Ω does not have calibre α.

Proof. Since β is a strong limit cardinal and $\beta^+ = 2^\beta$, there is by Theorem 6.18 an extremally disconnected, compact Hausdorff space Ω such that

Ω has a strictly positive measure (and hence satisfies c.c.c.), and

Ω does not have calibre β^+.

That Ω does not have calibre α follows from Theorem 2.3(a).

6.20 *Remarks.* (a) Corollary 6.19 shows that, at least assuming the generalized continuum hypothesis, the result of Corollary 5.26 is best possible.

(b) For $\alpha = \omega^+$, $\beta = \omega$ the space Ω of Corollary 6.19 is a compact c.c.c. space that does not have calibre ω^+. In Theorems 7.9 and 7.13 below, again assuming $\omega^+ = 2^\omega$, we define two more such spaces. We note (from Corollary 6.17) that in fact the spaces of Corollary 6.19 have property $\mathrm{K}_{\alpha,n}$ for all natural numbers $n \geq 2$ and all cardinals α with $\mathrm{cf}(\alpha) > \omega$.

(c) It follows from Theorem 6.18 that, at least assuming the generalized continuum hypothesis, for arbitrarily large regular cardinals α there are extremally disconnected compact Hausdorff spaces with c.c.c. that fail to have calibre α.

6.21 *Theorem.* Let $\omega^+ \leqslant \alpha$ and $\omega^+ \leqslant \mathrm{cf}(\alpha)$ and let X be a compact Hausdorff space with a strictly positive regular Borel measure μ. If $\{S_\xi : \xi < \alpha\}$ is a set of μ-measurable subsets of X with $\mu(S_\xi) > 0$ for $\xi < \alpha$, then there is $A \subset \alpha$ with $|A| = \alpha$ such that $\cap_{\xi \in A} S_\xi \neq \varnothing$.

Proof. Let $\mathscr{B}$ be the σ-algebra of Borel sets in X and let Ω be the Stone space of the measure algebra of $(X, \mathscr{B}, \mu)$. It follows from Theorem C.6 of the Appendix that Ω has a strictly positive probability measure (and hence has the c.c.c.); and from Theorem 5.6 it then follows that the (compact) space Ω has calibre α.

Since μ and its Carathéodory completion (cf. Theorem C.3 of the Appendix) are regular Borel measures there is for $\xi < \alpha$ a compact set $K_\xi \subset S_\xi$ such that $\mu(K_\xi) > 0$. The μ-equivalence class $[K_\xi]$ corresponds via Stone's duality to a non-empty, open-and-closed subset of Ω, and since Ω has calibre α there is $A \subset \alpha$ with $|A| = \alpha$ such that

$$\cap\{[K_\xi] : \xi \in A\} \neq \varnothing.$$

Since $\{[K_\xi] : \xi \in A\}$ has the finite intersection property in the measure algebra, the family $\{K_\xi : \xi \in A\}$ has the finite intersection property in X and we have from compactness that

$$\bigcap_{\xi \in A} S_\xi \supset \bigcap_{\xi \in A} K_\xi \neq \varnothing,$$

as required.

If instead of a strictly positive measure we assume only the weaker (ω, ω)-chain condition (defined below), we can nevertheless establish a chain condition of calibre type, namely property (K) (cf. Lemma 6.22(b)); that the stronger properties K_n for $2 < n < \omega$ need not hold in this context will follow, at least assuming the continuum hypothesis, from Argyros' example below (Theorem 6.25).

Definition. Let κ be a cardinal and X a space. Then X satisfies the (κ, κ)-*chain condition* if the set $\mathscr{T}^*(X)$ of non-empty, open subsets of X can be written in the form

$$\mathscr{T}^*(X) = \bigcup_{i < \kappa} \mathscr{T}_i$$

where, for $i < \kappa$, no κ elements of the set $\mathscr{T}_i$ are pairwise disjoint.

6.22 *Lemma.* Let X be a space.

(a) If X has property $(*)$ then X satisfies the (ω, ω)-chain condition.

(b) If X satisfies the (ω, ω)-chain condition then X has property (K).

Proof. (a) is obvious. We prove (b). Let $\mathscr{T}^*(X) = \cup_{n<\omega} \mathscr{T}_n$ where, for $n < \omega$, the set $\mathscr{T}_n$ has no infinite, pairwise disjoint subfamily. Let $\{U_\xi : \xi < \omega^+\} \subset \mathscr{T}^*(X)$. There are $\bar{n} < \omega$ and $A \subset \omega^+$ with $|A| = \omega^+$ such that $\{U_\xi : \xi \in A\} \subset \mathscr{T}_{\bar{n}}$. For $\{\xi, \xi'\} \in [A]^2$ we write

$$\begin{aligned} \{\xi, \xi'\} &\in P_0 \quad \text{if } U_\xi \cap U_{\xi'} \neq \varnothing \\ &\in P_1 \quad \text{if } U_\xi \cap U_{\xi'} = \varnothing. \end{aligned}$$

Since $[A]^2 = P_0 \cup P_1$ and $\omega \ll \omega^+$, it follows from Theorem 1.7 that there is $B \subset A$ such that either

$$\begin{aligned} &\text{type } B = \omega + 1 \quad \text{and } [B]^2 \subset P_1, \text{ or} \\ &|B| = \omega^+ \quad \text{and } [B]^2 \subset P_0. \end{aligned}$$

Since $\mathscr{T}_{\bar{n}}$ has no infinite, pairwise disjoint subfamily the first possibility cannot occur; hence $B \subset A \subset \omega^+$ and $|B| = \omega^+$ and

$$U_\xi \cap U_{\xi'} \neq \varnothing \quad \text{for } \xi, \xi' \in B,$$

as required.

Gaifman's example

6.23 *Theorem.* There is an extremally disconnected, compact Hausdorff space X such that

(a) X has property $(*)$ (hence, X is a c.c.c. space),

(b) X does not have property $(**)$ (hence, X does not have a strictly positive measure),

(c) X has property K_n for $2 \leq n < \omega$, and

(d) assuming the continuum hypothesis, X does not have calibre ω^+.

Proof. It follows from Lemma 6.2 and Corollary 2.22 that it is enough to find a (compact, Hausdorff) space X with properties (a), (b), (c) and (d); for then the Gleason space of X will have all the required properties.

Let $\{T_n : 2 \le n < \omega\}$ be an enumeration of the set of open intervals in $\mathbb{R}$ with rational endpoints, for $2 < n < \omega$ let

$$\{T_{n,k} : 1 \le k \le n^2\}$$

be a family of subintervals of T_n with rational endpoints such that

$$T_{n,k} \cap T_{n,k}{}' = \varnothing \text{ for } 1 \le k < k' \le n^2,$$

and set

$$X = \{p \in \{0,1\}^{\mathbb{R}} : |\{k : 1 \le k \le n^2, p|T_{n,k} \not\equiv 0\}| < n \quad \text{for } 2 \le n < \omega\}.$$

For every basic open subset $U = \prod_{t \in \mathbb{R}} U_t$ of $\{0, 1\}^{\mathbb{R}}$ we denote as usual by $R(U)$ the (finite) restriction set $R(U)$ of U:

$$R(U) = \{t \in \mathbb{R} : U_t \ne \{0, 1\}\}.$$

We set

$$R^0(U) = \{t \in R(U) : \pi_t[U] = \{0\}\} \quad \text{and}$$
$$R^1(U) = \{t \in R(U) : \pi_t[U] = \{1\}\},$$

for $2 \le n < \omega$ we set

$$R^1(U, n) = \{k : 1 \le k \le n^2, R^1(U) \cap T_{n,k} \ne \varnothing\},$$

and we note that

$$X = \{0, 1\}^{\mathbb{R}} \backslash \cup \{U : U \text{ is basic open}, |R^1(U, n)| \ge n \text{ for some } n \ge 2\}.$$

Thus X is a compact Hausdorff space.

Now for $2 \le n < \omega$, $1 \le m < \omega$ we set

$$\mathscr{B}_{n,m} = \{U : U \text{ is basic open in } \{0, 1\}^{\mathbb{R}}, |R(U)| \le n/m\},$$

and we claim that if $\{U_i : 1 \le i \le m\} \subset \mathscr{B}_{n,m}$ with

$$U_i \cap X \ne \varnothing \quad \text{for } 1 \le i \le m,$$
$$\bigcap_{i=1}^{m} U_i \ne \varnothing, \quad \text{and}$$
$$R^1(U_i, k) = R^1(U_{i'}, k) \quad \text{for } 1 \le i \le i' \le m, 2 \le k \le n,$$

then $(\cap_{i=1}^m U_i) \cap X \ne \varnothing$.

Indeed since $\cap_{i=1}^m U_i \ne \varnothing$ we have

$$\left(\bigcup_{i=1}^{m} R^0(U_i)\right) \cap \left(\bigcup_{i=1}^{m} R^1(U_i)\right) = \varnothing;$$

we define $p \in \{0, 1\}^{\mathbb{R}}$ by

$$p_t = 1 \quad \text{if } t \in \bigcup_{i=1}^{m} R^1(U_i)$$
$$= 0 \quad \text{otherwise.}$$

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We have $p \in \cap_{i=1}^m U_i$; we verify that $p \in X$. For $2 \leq j < \omega$, set

$$I(j) = \{k: 1 \leq k \leq j^2, p|T_{j,k} \not\equiv 0\}.$$

Then for $n < j$ we have

$$|I(j)| \leq |\{t \in \mathbb{R}: p_t = 1\}| = \left| \bigcup_{i=1}^m R^1(U_i) \right|$$
$$\leq mn/m = n < j,$$

and for $2 < j \leq n$ we have

$$|I(j)| = \left| \bigcup_{i=1}^m R^1(U_i, j) \right| = |R^1(U_1, j)| < j$$

(the inequality since $U_1 \cap X \neq \emptyset$). It follows that $p \in X$, and the claim is proved.

For $2 \leq n < \omega$ and $J_k \subset \{1, 2, \ldots, k^2\}$ for $2 \leq k \leq n$ we set $\mathscr{B}_{n,m}(J_2, \ldots, J_n) = \{U \in \mathscr{B}_{n,m}: R^1(U, k) = J_k \text{ for } 2 \leq k \leq n\}$ for $1 \leq m < \omega$; and we define

$$s(n) = \sum_{2 \leq k \leq n^2} 2^{(k^2)}, \quad p_n = 2^{n/2} \cdot 2^{s(n)}$$

We verify statements (a), (b), (c) and (d).

(a) From (the case $m = 2$ of) the claim just proved it follows that if $\mathscr{F} \subset \mathscr{B}_{n,2}$ with

$$U \cap X \neq \emptyset \quad \text{for } U \in \mathscr{F},$$
$$U \cap V \cap X = \emptyset \quad \text{for } U, V \in \mathscr{F}, U \neq V, \quad \text{and}$$
$$R^1(U, k) = R^1(V, k) \text{ for } U, V \in \mathscr{F}, 2 \leq k \leq n,$$

then the elements of $\mathscr{F}$ are pairwise disjoint. Since the (product) measure μ on $\{0, 1\}^{\mathbb{R}}$ determined by the coordinate measures

$$\mu_t(\{0\}) = \mu_t(\{1\}) = 1/2$$

satisfies $\mu(\{0, 1\}^{\mathbb{R}}) = 1$ and $\mu(U) \geq (1/2)^{n/2}$ for $U \in \mathscr{B}_{n,2}$, we have $|\mathscr{F}| \leq 2^{n/2}$.

We note that if $\mathscr{F} \subset \mathscr{B}_{n,2}$ with

$$U \cap X \neq \emptyset \quad \text{for } U \in \mathscr{F} \text{ and}$$
$$U \cap V \cap X = \emptyset \quad \text{for } U, V \in \mathscr{F}, U \neq V,$$

then

$$|\mathscr{F}| \leq \sum \{|\mathscr{F} \cap \mathscr{B}_{n,2}(J_2, \ldots, J_n)| : J_k \subset \{1, \ldots, k^2\}, 2 \leq k \leq n\} \leq p_n.$$

Finally for $2 \leq n < \omega$ set

$$\mathscr{T}_n = \{W \in \mathscr{T}^*(X)\text{: there is } U \in \mathscr{B}_{n,2} \quad \text{such that } W \supset U \cap X \neq \varnothing\}.$$

Then

$$\mathscr{T}^*(X) = \cup_{2 \leq n < \omega} \mathscr{T}_n, \quad \text{and}$$

no $p_n + 1$ elements of the family $\mathscr{T}_n$ are pairwise disjoint.

It then follows that X has property (*), as required.

(b) We show that X does not have property (**), i.e., that if

$$\mathscr{T}^*(X) = \bigcup_{n<\omega} \mathscr{S}_n$$

then there is $\bar{n} < \omega$ such that $\kappa(\mathscr{S}_{\bar{n}}) = 0$.

For $t \in \mathbb{R}$ we have $\pi_t^{-1}(\{1\}) \cap X \neq \varnothing$. We set

$$I_n = \{t \in \mathbb{R} : \pi_t^{-1}(\{1\}) \cap X \in \mathscr{S}_n\} \quad \text{for } n < \omega$$

and we note from the Baire category theorem that not all of the sets I_n are nowhere dense; that is, there are $\bar{n} < \omega$ and a non-empty interval $T \subset \mathbb{R}$ such that

$$I_{\bar{n}} \cap T \text{ is dense in } T.$$

We set

$$A = \{n < \omega\colon T_n \subset T\},$$

we choose

$$t_{n,k} \in I_{\bar{n}} \cap T_{n,k} \quad \text{for } n \in A,\, 1 \leq k \leq n^2,$$

and we set

$$\mathscr{F}_n = \{\pi_{t_{n,k}}^{-1}(\{1\}) \cap X : 1 \leq k \leq n^2\} \quad \text{for } n \in A.$$

Then $\mathscr{F}_n \subset \mathscr{S}_{\bar{n}}$ for $n \in A$; since no n elements of $\mathscr{F}_n$ have non-empty intersection we have

$$\operatorname{cal} \mathscr{F}_n \leq n - 1$$

and hence

$$\kappa(\mathscr{S}_{\bar{n}}) \leq (\operatorname{cal} \mathscr{F}_n)/|\mathscr{F}_n| \leq (n-1)/n^2 < 1/n \quad \text{for } n \in A.$$

Since $|A| = \omega$ we have $\kappa(\mathscr{S}_{\bar{n}}) = 0$, as required.

(c) We fix m such that $1 \leq m < \omega$ and we show that X has property K_m. Let $\{U_\xi : \xi < \omega^+\}$ be a set of basic open subsets of $\{0, 1\}^{\mathbb{R}}$ with $U_\xi \cap X \neq \varnothing$ for $\xi < \omega^+$. Since $\{0, 1\}^{\mathbb{R}}$ has calibre ω^+ (cf. Corollary 3.7(a)), we assume without loss of generality that $\cap_{\xi<\omega^+} U_\xi \neq \varnothing$.

We assume further without loss of generality, passing if necessary to a subfamily $\{U_\xi : \xi \in I\}$ with $I \subset \omega^+$ and $|I| = \omega^+$, that there are n with $2 \leq n < \omega$ and $J_2, \ldots, J_n \subset \{1, 2 \ldots, n^2\}$ such that

$$\{U_\xi : \xi < \omega^+\} \subset \mathscr{B}_{n,m}(J_2, \ldots, J_n).$$

It then follows from the claim proved above that for $A \subset \omega^+$ with $|A| = m$ we have $(\cap_{\xi \in A} U_\xi) \cap X \neq \emptyset$, as required.

(d) Assume the continuum hypothesis, and let $S = \{x_\xi : \xi < \omega^+\}$ be a faithfully indexed Lusin set of $\mathbb{R}$ (cf. Theorem B.4 of the Appendix). We consider $\{\pi_{x_\xi}^{-1}(\{1\}) \cap X : \xi < \omega^+\}$, a family of non-empty, open subsets of X. Suppose there are $A \subset \omega^+$ with $|A| = \omega^+$ and $p \in \{0, 1\}^{\mathbb{R}}$ such that

$$p \in \bigcap_{\xi \in A} \pi_{x_\xi}^{-1}(\{1\}) \cap X.$$

Since S is of Category II in $\mathbb{R}$ there is n with $2 \leq n < \omega$ such that $S \cap T_n$ is dense in T_n (and hence $S \cap T_{n,k} \neq \emptyset$ for $1 \leq k \leq n^2$). It follows that

$$|\{k : 1 \leq k \leq n^2, p | T_{n,k} \not\equiv 0\}| = n^2 > n,$$

contradicting the condition $p \in X$.

The proof of the theorem is complete.

6.24 *Problem.* Is there a compact Hausdorff space X such that X has property (*), X does not have a strictly positive measure, and X has calibre ω^+?

We think that, even without special set-theoretic assumptions, it should be possible to define such a space.

The example of Argyros

6.25 *Theorem* (Argyros). There is an extremally disconnected, compact Hausdorff space X such that

(a) X has property (*) (and hence property (K));

(b) X does not have property (**) (hence, X does not have a strictly positive measure); and

(c) assuming the continuum hypothesis, X does not have property K_3.

Proof. It is enough to define a (completely regular, Hausdorff) space X satisfying (a), (b) and (c). For then, according to Theorem 2.17(d) and Lemma 6.2(c), (d), the Gleason space of X will have all the required properties.

We begin by defining a tree $\langle T, \preccurlyeq \rangle$ with

$$T = \cup_{n<\omega} T_n \subset [\omega]^2.$$

Let $\{S_{n,j}: n<\omega, 1 \leq j \leq 3^n\}$ be a family such that

$$S_{n,j} \in [\omega]^3 \quad \text{and}$$
$$S_{n,j} \cap S_{n',j'} = \varnothing \quad \text{for } \langle n,j \rangle \neq \langle n',j' \rangle,$$

set $T_n = \cup \{[S_{n,j}]^2 : 1 \leq j \leq 3^n\}$ for $n<\omega$ (so that $|T_n| = 3^{n+1}$), let

$$T_n = \{s_j : 1 \leq j \leq 3^{n+1}\}$$

be an enumeration of T_n, and set $T = \cup_{n<\omega} T_n$. The order $\preccurlyeq$ of T is defined so that the (immediate) successors of elements of T_n are in T_{n+1}, and is determined by the following rule: if $s \in T_n$ and $t \in T_{n+1}$, then $s \prec t$ if and only if there is j with $1 \leq j \leq 3^{n+1}$ such that $s = s_j$ and $t \in [S_{n+1,j}]^2$.

We set

$$A(n) = \cup \{s \in T : s \in \cup_{n \leq p < \omega} T_p\} \quad \text{and}$$
$$B(n) = \omega \backslash A(n) \quad \text{for } n < \omega.$$

For $s \in T$ we denote by K_s the set (of cardinality 2) of non-constant functions in $\{0, 1\}^s$; that is, if $s = \{k, l\}$ then

$$K_s = \{\{\langle k, 1 \rangle, \langle l, 0 \rangle\}, \{\langle k, 0 \rangle, \langle l, 1 \rangle\}\}.$$

We note that if $S \subset \omega$ with $|S| = 3$ and if there is $n < \omega$ such that

$$[S]^2 = \{s_1, s_2, s_3\} \subset T_n,$$

then the family $\{K_{s_q} \times \{0,1\}^{\omega \backslash s_q} : 1 \leq q \leq 3\}$, which has three elements, has empty intersection and any two elements of the family have non-empty intersection.

We denote by $\boldsymbol{\Sigma}$ the set of (finite or infinite) branches of T, and for $\Sigma \in \boldsymbol{\Sigma}$ we set

$$A(\Sigma) = \cup \{s : s \in \Sigma\} \quad \text{and}$$
$$V_\Sigma = \left(\prod_{s \in \Sigma} K_s\right) \times \{0, 1\}^{\omega \backslash A(\Sigma)}.$$

We now define the space $\langle X, \mathscr{T} \rangle$. As a set, X is the Cantor set $\{0, 1\}^\omega$; the topology $\mathscr{T}$ is defined by the subbase that contains

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(i) all subsets of $\{0,1\}^\omega$ that are open-and-closed in the (usual) product topology of $\{0,1\}^\omega$, and

(ii) all sets of the form V_Σ for $\Sigma \in \mathbf{\Sigma}$.

The base $\mathscr{B}$ determined by this subbase consists of all sets of the form $U \cap (\cap_{j=1}^m V_{\Sigma_j})$ with U open-and-closed in the usual topology and with $\{\Sigma_j : 1 \le j \le m\} \subset \mathbf{\Sigma}$. The elements of $\mathscr{B}$ are closed in the usual topology of $\{0,1\}^\omega$, hence in its extension $\mathscr{T}$; thus $\langle X, \mathscr{T} \rangle$ is a completely regular, Hausdorff space.

We say that the basic set $V = U \cap (\cap_{j=1}^m V_{\Sigma_j})$ is *determined by* $(U; \Sigma_1, \ldots, \Sigma_m)$, and we say that V is *separated at level n* if

(i) $(\Sigma_{j_1} \cap T_n) \cap (\Sigma_{j_2} \cap T_n) = \varnothing$ for $1 \le j_1 < j_2 \le m$, and

(ii) there is a finite subset F of ω on which U depends such that $F \cap A(n) = \varnothing$.

If $V = U \cap (\cap_{j=1}^m V_{\Sigma_j}) \in \mathscr{B}$ and V is separated at level n we set

$$V(0,n) = \pi_{B(n)}[V] \quad \text{and}$$

$$V(1,n) = \bigcap_{j=1}^{m} \left(\left(\prod_{s \in \Sigma_j,\, s \subset A(n)} K_s \right) \times \{0,1\}^{A(n) \setminus A(\Sigma_j)} \right);$$

then $V(0,n)$ is open-and-closed in the usual topology of $\{0,1\}^{B(n)}$, and $V = V(0,n) \times V(1,n)$.

For use in the sequel we verify the following claim:

Statement (1). Let $\{V_i : i \in I\} \subset \mathscr{B}$ with V_i determined by $(U^i; \Sigma_1^i, \ldots, \Sigma_{m(i)}^i)$ and suppose there is $n < \omega$ such that each V_i is separated at level n. Then the following are equivalent:

($\tilde{a}$) $\cap_{i \in I} V_i \neq \varnothing$; and

($\tilde{b}$)1. $\cap_{i \in I} V_i(0,n) \neq \varnothing$, and

2. of any three distinct elements of the set $\{\Sigma_j^i \cap T_p : n \le p < \omega,\ i \in I,\ 1 \le j \le m(i)\}$, some two have empty intersection.

It is clear that ($\tilde{a}$) $\Rightarrow$ ($\tilde{b}$)1. If ($\tilde{a}$) holds and $\{s_q : 1 \le q \le 3\}$ are distinct elements of the indicated set with

$$s_q = \Sigma_{j(q)}^{i(q)} \cap T_{p(q)} \quad \text{for } 1 \le q \le 3,$$

no two with empty intersection, then there is $p < \omega$ such that each natural number $p(q)$ is equal to p and there is $S \subset \omega$ with $|S| = 3$ such that

$$[S]^2 = \{s_q : 1 \le q \le 3\} \subset T_p;$$

we have

$$\bigcap_{i\in I} V_i \subset \bigcap_{q=1}^{3} V_{i(q)} \subset \bigcap_{q=1}^{3} K_{s_q} \times \{0,1\}^{\omega\setminus s_q} = \varnothing$$

a contradiction.

We prove $(\tilde{\mathrm{b}}) \Rightarrow (\tilde{\mathrm{a}})$. Since $\cap_{i\in I} V_i(0,n) \neq \varnothing$ by $(\tilde{\mathrm{b}})1$, it is enough to show $\cap_{i\in I} V_i(1,n) \neq \varnothing$. We note (from the definition of T) that if $s_1, s_2 \in T$ with $s_1 \cap s_2 \neq \varnothing$ then there is $p < \omega$ such that $s_1, s_2 \in T_p$; and if $s_1, s_2, s_3 \in T$ with $s_1 \cap s_2 \neq \varnothing$ and $s_1 \cap s_3 \neq \varnothing$ then $s_2 \cap s_3 \neq \varnothing$. It then follows from $(\tilde{\mathrm{b}})2$ that there is a sequence $\{B_r : r < \omega\}$ such that

$$\bigcup_{r<\omega} B_r = \{\Sigma_j^i \cap T_p : n \le p < \omega, i \in I, 1 \le j \le m(i)\},$$

$$1 \le |B_r| \le 2 \quad \text{for } r < \omega,$$

$$B_r \cap B_{r'} = \varnothing \quad \text{for } r < r' < \omega, \quad \text{and}$$

$$\text{if } s \in B_r, s' \in B_{r'} \text{ with } r < r' < \omega \text{ then } s \cap s' = \varnothing.$$

For $r < \omega$ there is $x(r) \in \cap_{s\in B_r} K_s$, and there is $x \in \{0,1\}^{A(n)}$ such that $x_s = x(r)_s$ for $s \in B_r$. It is then clear that $x \in \cap_{i\in I} V(1,n)$, as required.

The proof of Statement (1) is complete.

We note a consequence of Statement (1).

Statement (2). Let $\{V_i : i \in I\} \subset \mathscr{B}$ and suppose there is $n < \omega$ such that each V_i is separated at level n. If $\cap_{i\in I} V(0,n) \neq \varnothing$, then $\cap_{i\in I} V_i \neq \varnothing$ if and only if every three elements of the set $\{V_i : i \in I\}$ have non-empty intersection.

Finally we verify that the space $\langle X, \mathscr{T} \rangle$ satisfies (a), (b) and (c).

(a) For $n < \omega$ let μ denote the usual Haar measure on $\{0,1\}^{B(n)}$ and set

$$\mathscr{T}_{n,m} = \{U \in \mathscr{T}^*(X) : \text{there is } V \in \mathscr{B} \text{ such that } V \subset U, V \text{ is separated at level } n, \text{ and } (\mu(V(0,n)) \ge 1/(m+1)\}.$$

If $\mathscr{U} \subset \mathscr{T}_{n,m}$ with $|\mathscr{U}| = m+2$ then there are $U_0, U_1 \in \mathscr{U}$ and $V_0, V_1 \in \mathscr{B}$ such that

$$V_0 \subset U_0, V_1 \subset U_1,$$

$$V_0 \text{ and } V_1 \text{ are separated at level } n, \text{ and}$$

$$V_0(0,n) \cap V_1(0,n) \neq \varnothing.$$

Since each branch of T determining V_0 intersects at most one branch of T determining V_1 at a level greater than or equal to n, the family $\{V_0, V_1\}$ satisfies (not only condition $(\tilde{\mathrm{b}})1$ but also) condition $(\tilde{\mathrm{b}})2$ of Statement (1); hence

$$U_0 \cap U_1 \supset V_0 \cap V_1 \neq \varnothing.$$

It follows that no $m+2$ elements of $\mathscr{T}_{n,m}$ are pairwise disjoint. That X has property (*) now follows easily.

(b) If X has property (**) then the set $\mathscr{T}^*(X)$ can be written in the form

$$\mathscr{T}^*(X) = \bigcup_{n<\omega} \mathscr{T}'_n$$

with $\kappa(\mathscr{T}'_n) > 0$ for $n > \omega$. We set

$$\mathscr{T}_n = \{U \in \mathscr{T}^*(X)\colon \text{there is } V \subset U \text{ such that } V \in \mathscr{T}'_n\},$$

and we note that

$$\mathscr{T}^*(X) = \bigcup_{n<\omega} \mathscr{T}_n,$$
$$\kappa(\mathscr{T}_n) > 0 \quad \text{for } n < \omega, \text{ and}$$
$$\text{if } U, V \in \mathscr{T}^*(X) \quad \text{with } V \subset U, \text{ then}$$
$$\min\{n : U \in \mathscr{T}_n\} \leq \min\{n : V \in \mathscr{T}_n\}.$$

We claim there are $\bar{n} < \omega$ and a finite branch $\bar{\Sigma}$ of T such that if Σ is a finite branch and $\bar{\Sigma} \subset \Sigma$ then $V_\Sigma \in \mathscr{T}_{\bar{n}}$. If the claim fails there are a sequence $\{n_k : k < \omega\}$ of natural numbers and a sequence $\{\Sigma_k : k < \omega\}$ of finite branches such that

$$n_k < n_{k+1} \text{ and } \Sigma_k \subset \Sigma_{k+1} \text{ for } k < \omega, \text{ and}$$
$$\min\{n : V_{\Sigma_k} \in \mathscr{T}_n\} = n_k \quad \text{for } k < \omega.$$

We set $\Sigma = \cup_{k<\omega} \Sigma_k$ and we choose $n < \omega$ such that $V_\Sigma \in \mathscr{T}_n$. Since $\Sigma_k \subset \Sigma$ we have $V_\Sigma \subset V_{\Sigma_k}$ and hence $V_{\Sigma k} \in \mathscr{T}_n$ for all $k < \omega$, a contradiction The proof of the claim is complete. We choose $\bar{n}$ and $\bar{\Sigma}$ as given and we note by m the length of $\bar{\Sigma}$.

We choose $k < \omega$ such that $(2/3)^k < \kappa(\mathscr{T}_{\bar{n}})$ and we set

$$\mathscr{S} = \{V_\Sigma : \bar{\Sigma} \subset \Sigma, \Sigma \text{ is finite, length of } \Sigma \text{ is } m+k\}.$$

We note that $\mathscr{S} \subset \mathscr{T}_{\bar{n}}$ and $|\mathscr{S}| = 3^k$. We note further that if $\mathscr{U} \subset \mathscr{S}$ with $|\mathscr{U}| \geq 2^k + 1$ then there are $V_{\Sigma(1)}, V_{\Sigma(2)}, V_{\Sigma(3)} \in \mathscr{U}$ and p with $m \leq p < m + k$ such that

$$\Sigma(1) \cap T_p = \Sigma(2) \cap T_p = \Sigma(3) \cap T_p \text{ and}$$
$$\text{the sets } \Sigma(1) \cap T_{p+1}, \Sigma(2) \cap T_{p+1}, \Sigma(3) \cap T_{p+1} \text{ are distinct.}$$

No two of the sets $\Sigma(1)\cap T_{p+1}$, $\Sigma(2)\cap T_{p+1}$, $\Sigma(3)\cap T_{p+1}$ have empty intersection, so from Statement (1) we have

$$\cap\mathscr{U}\subset V_{\Sigma(1)}\cap V_{\Sigma(2)}\cap V_{\Sigma(3)}=\varnothing.$$

It follows that cal $\mathscr{S}\leq 2^k$ and hence

$$\kappa(\mathscr{T}_{\bar{n}})\leq \text{cal}\,\mathscr{S}/3^k\leq(2/3)^k,$$

a contradiction.

(c) We note from (b) that X is not a separable space (cf. Corollary 6.7 and Lemma 6.1(a)). It follows from the continuum hypothesis that

$$|X|=d(X)=\omega^+,$$

and then from Lemma 5.27 (with $\mathscr{T}_1=\mathscr{T}$ and with $\mathscr{T}_2$ the usual topology of $\{0,1\}^\omega$) it follows that the space $\langle X,\mathscr{T}\rangle$ does not have calibre ω^+. There is $\{V_\xi:\xi<\omega^+\}\subset\mathscr{B}$ such that if $I\subset\omega^+$ with $|I|=\omega^+$ then $\cap_{\xi\in I}V_\xi=\varnothing$. There are $n<\omega$ and $A\subset\omega^+$ with $|A|=\omega^+$ such that each of the sets V_ξ with $\xi\in A$ is separated at level n. We assume without loss of generality, using the fact that the number of open-and-closed subsets of $\{0,1\}^{B(n)}$ is ω, that there is a (fixed) open-and-closed subset $V(0,n)$ of $\{0,1\}^{B(n)}$ such that

$$V_\xi(0,n)=V(0,n)\quad\text{for all } \xi\in A.$$

For $I\subset A$ with $|I|=\omega^+$ we then have

$$\bigcap_{\xi\in I}V_\xi(0,n)=V(0,n)\neq\varnothing,$$

and since $\cap_{\xi\in I}V_\xi=\varnothing$ it then follows from Statement (2) that some three elements of the set $\{V_\xi:\xi\in I\}$ have empty intersection. Thus X does not have property K_3.

The proof is complete.

We indicate how the method of (the proof of) Theorem 6.25 can be used to prove a more general result.

6.26 *Theorem* (Argyros). Let $2\leq m<\omega$. There is an extremally disconnected, compact Hausdorff space X such that

(a) X has property (*) and property K_m;

(b) X does not have property (**) (hence, X does not have a strictly positive measure); and

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(c) assuming the continuum hypothesis, X does not have property K_{m+1}.

Proof. (We note that Theorem 6.25 is the case $m = 2$.) As before, it is enough to find a (completely regular, Hausdorff) space X satisfying (a), (b) and (c).

We fix a natural number p and a family $\{F_i : 1 \le i \le m+1\}$ of subsets of $\{0, 1\}^p$ such that

$$\bigcap_{i=1}^{m+1} F_i = \emptyset \quad \text{and}$$
$$\bigcap_{\substack{i=1\\ i\neq j}}^{m+1} F_i \neq \emptyset \quad \text{for } 1 \le j \le m+1.$$

Let $\{S_{n,j} : n < \omega, 1 \le j \le (m+1)^n\}$ be a family such that

$$S_{n,j} \in [\omega]^p \quad \text{and}$$
$$S_{n,j} \cap S_{n',j'} = \emptyset \quad \text{for } \langle n, j\rangle \neq \langle n', j'\rangle,$$

let $\varphi_{n,j}$ be a one-to-one function from $S_{n,j}$ onto p, and set

$$F_{i,n,j} = \{t \circ \varphi_{n,j} : t \in F_i\} \quad \text{for } 1 \le i \le m+1, \text{and}$$
$$\mathscr{F}_{n,j} = \{F_{i,n,j} : 1 \le i \le m+1\} \quad \text{for } n < \omega, 1 \le j \le (m+1)^n.$$

We note that $\mathscr{F}_{n,j}$ is a family of subsets of $\{0, 1\}^{S_{n,j}}$ such that $\cap \mathscr{F}_{n,j} = \emptyset$ and such that if $\mathscr{G} \subset \mathscr{F}_{n,j}$ with $|\mathscr{G}| \le m$ then $\cap \mathscr{G} \neq \emptyset$.

We set

$$T_n = \bigcup_{j=1}^{(m+1)^n} \mathscr{F}_{n,j} \quad \text{for } n < \omega,$$

we note that $|T_n| = (m+1)^{n+1}$, we let

$$T_n = \{s_j : 1 \le j \le (m+1)^{n+1}\}$$

be an enumeration of T_n, and we set $T = \cup_{n<\omega} T_n$. The order $\preccurlyeq$ of the tree T is defined so that the (immediate) successors of elements of T_n are in T_{n+1}, and is determined by the following rule: if $s \in T_n$ and $t \in T_{n+1}$, then $s \prec t$ if and only if there is j with $1 \le j \le (m+1)^{n+1}$ such that $s = s_j$ and $t \in \mathscr{F}_{n+1,j}$.

Again we denote by $\mathbf{\Sigma}$ the set of all branches of T, and for $\Sigma \in \mathbf{\Sigma}$ we set

$$V_\Sigma = \left(\prod_{F \in \Sigma} F\right) \times \{0, 1\}^{\omega \setminus \cup \Sigma}.$$

We now define the space $\langle X, \mathscr{T} \rangle$. As a set, X is the Cantor set $\{0, 1\}^\omega$; the topology $\mathscr{T}$ is defined by the subbase that contains

(i) all subsets of $\{0, 1\}^\omega$ that are open-and-closed in the (usual) product topology of $\{0, 1\}^\omega$, and

(ii) all sets of the form V_Σ for $\Sigma \in \mathbf{\Sigma}$.

The verification that the space $\langle X, \mathscr{T} \rangle$ is as required is analogous to the case $m = 2$ (Theorem 6.25), and is left to the reader.

6.27 *Problem* (Argyros and Negrepontis). Assume the generalized continuum hypothesis. Let $\omega^+ < \alpha = \beta^+$ with $\mathrm{cf}(\beta) = \omega$ and let $2 \le m < \omega$. Is there a c.c.c. space X such that X has property $\mathrm{K}_{\alpha,m}$ and X does not have property $\mathrm{K}_{\alpha,m+1}$?

In the Notes to this chapter we outline a proof that if the requirement that X be a c.c.c. space is omitted then such a space exists; in this case the assumption $\mathrm{cf}(\beta) = \omega$ may be omitted.

6.28 *Lemma*. Let β be an infinite cardinal and X a compact Hausdorff space. The following statements are equivalent.

(a) The set $\mathscr{T}^*(X)$ can be written in the form $\mathscr{T}^*(X) = \cup_{i<\beta} \mathscr{T}_i$ with $\kappa(\mathscr{T}_i) > 0$ for $i < \beta$; and

(b) there is a set $\{\mu_i : i < \beta\}$ of regular, Borel probability measures on X such that for every $U \in \mathscr{T}^*(X)$ there is $i < \beta$ with $\mu_i(U) > 0$.

Proof. (a) $\Rightarrow$ (b). Let $\{\mathscr{T}_i : i < \beta\}$ be as in (a) and for $i < \beta$ let μ_i be a regular, Borel probability measure such that

$$\mu_i(\mathrm{cl}_X V) \ge \kappa(\mathscr{T}_i) \quad \text{for } V \in \mathscr{T}_i.$$

For $U \in \mathscr{T}^*(X)$ there is $V \in \mathscr{T}^*(X)$ such that $\mathrm{cl}_X V \subset U$, and if $V \in \mathscr{T}_i$ we have

$$\mu_i(U) \ge \mu_i(\mathrm{cl}_X V) \ge \kappa(\mathscr{T}_i) > 0.$$

(b) $\Rightarrow$ (a). Let $\{\mu_i : i < \beta\}$ be as in (b) and for $i < \beta, n < \omega$ set

$$\mathscr{T}_{i,n} = \{U \in \mathscr{T}^*(X) : \mu_i(U) \ge 1/(n+1)\}.$$

Then $\mathscr{T}^*(X) = \cup_{i<\beta, n<\omega} \mathscr{T}_{i,n}$, and from Lemma 6.1 (with $\mathscr{A} = \mathscr{U} = \mathscr{T}_{i,n}$) we have

$$\kappa(\mathscr{T}_{i,n}) \ge 1/(n+1) > 0,$$

as required.

6.29 *Theorem* (Argyros). Let β be a strong limit cardinal with $\mathrm{cf}(\beta) = \omega$. There is an extremally disconnected, compact Hausdorff space X such that

(a) X has property (*), and

(b) the set $\mathscr{T}^*(X)$ of non-empty, open subsets of X cannot be written in the form $\mathscr{T}^*(X) = \cup_{i<\beta} \mathscr{T}_i$ with $\kappa(\mathscr{T}_i) > 0$ for $i < \beta$ (i.e., for every set $\{\mu_i : i < \beta\}$ of regular, Borel measures on X there is a non-empty, open subset U of X such that $\mu_i(U) = 0$ for $i < \beta$).

Proof. It is enough to find a (completely regular, Hausdorff) space X satisfying (a) and the first part of (b). For then the Gleason space of X is an extremally disconnected, compact Hausdorff space satisfying (a) and the first part of (b); the equivalence of the two parts of (b) (for compact Hausdorff spaces) is given by Lemma 6.28.

There is a sequence $\{\alpha_n : n < \omega\}$ such that $\omega \leq \alpha_n < \alpha_{n'} < \beta$ for $n < n' < \omega$, and $\sum_{n<\omega} \alpha_n = \beta$. We set

$$\beta_0 = (2^{\alpha_0})^+, \quad \text{and}$$

$$\beta_{n+1} = \max\{(2^{\alpha_n})^+, (2^{\beta_n})^+\} \quad \text{for } 0 < n < \omega,$$

and we note that

$$\beta_n \text{ is a regular cardinal for } n < \omega,$$

$$\beta_n < \beta_{n'} < \beta \quad \text{for } n < n' < \omega,$$

$$\sum_{n<\omega} \beta_n = \beta, \quad \text{and}$$

$$(\beta_n)^+ \ll \beta_{n'} \quad \text{for } n < n' < \omega.$$

Now exactly as in the proof of Theorem 5.28 we choose a family $\{B_n : n < \omega\}$ of (pairwise disjoint) subsets of β with $|B_n| = \beta_n$ for $n < \omega$, we choose a family $\{B_{n+1,i} : i < \beta_n\}$ of (pairwise disjoint) subsets of B_{n+1} with $|B_{n+1,i}| = \beta_{n+1}$ for $n < \omega$, $i < \beta_n$, we define a tree $T = \cup_{n<\omega} T_n$ of height ω with

$$T_0 = [B_0]^2 \quad \text{and}$$

$$T_{n+1} = \cup\{[B_{n+1,i}]^2 : i < \beta_n\} \quad \text{for } n < \omega,$$

we denote by $\mathbf{\Sigma}$ the set of all branches of T, we define K_s for $s \in T$ and V_Σ for $\Sigma \in \mathbf{\Sigma}$ and we define a space $(X, \mathscr{T})$. (As a set, X is the set $\{0, 1\}^\beta$. The topology $\mathscr{T}$ is defined by the subbase that contains

(i) all subsets of $\{0, 1\}^\beta$ of the form $\pi_J^{-1}(S) \times \{0, 1\}^{\beta \setminus J}$ with $|J| < \omega$ and with S compact in the usual product topology of $\{0, 1\}^J$, and

(ii) all sets of the form V_Σ for $\Sigma \in \mathbf{\Sigma}$.

The proof that X has property (*) is entirely analogous to the proof of (a) in Theorem 6.25; we will not repeat the details here.

We prove (b). Suppose that $\mathscr{T}^*(X) = \cup_{i<\beta}\mathscr{T}_i$ with $\kappa(\mathscr{T}_i) > 0$ for $i < \beta$. We assume without loss of generality, replacing if necessary the set $\mathscr{T}_i$ by $\{U \in \mathscr{T}^*(X)$: there is $V \in \mathscr{T}_i$ such that $V \subset U\}$, that if $U, V \in \mathscr{T}^*(X)$ and $V \subset U$ then

$$\min\{i: U \in \mathscr{T}_i\} \leq \min\{i: V \in \mathscr{T}_i\}.$$

We claim that there are $\bar{n} < \omega$ and a finite branch $\bar{\Sigma}$ of T such that if Σ is a finite branch and $\bar{\Sigma} \subset \Sigma$ then there is $i < \beta_{\bar{n}}$ such that $V_\Sigma \in \mathscr{T}_i$. If the claim fails there are a sequence $\{n_k : k < \omega\}$ of natural numbers and a sequence $\{\Sigma_k : k < \omega\}$ of finite branches such that

$$n_k < n_{k+1} \text{ and } \Sigma_k \subset \Sigma_{k+1} \quad \text{for } k < \omega, \text{ and}$$

$$\min\{n : V_{\Sigma_k} \in \bigcup_{i<\beta_n} \mathscr{T}_i\} = n_k \quad \text{for } k < \omega.$$

We set $\Sigma = \cup_{k<\omega}\Sigma_k$ and we choose $n < \omega$ such that $V_\Sigma \in \cup_{i<\beta_n}\mathscr{T}_i$. Since $\Sigma_k \subset \Sigma$ we have $V_\Sigma \subset V_{\Sigma_k}$ and hence $V_{\Sigma_k} \in \cup_{i<\beta_n}\mathscr{T}_i$, a contradiction. We choose $\bar{n}$ and $\bar{\Sigma}$ as given, we set

$$S_m = \{i < \beta_{\bar{n}} : \kappa(\mathscr{T}_i) > 1/m\} \quad \text{for } 1 \leq m < \omega,$$

we note that $\beta_{\bar{n}} = \cup\{S_m : 1 \leq m < \omega\}$, and we claim that there are $\bar{m}$ with $1 \leq \bar{m} < \omega$ and a finite branch Σ' of T with $\Sigma \subset \Sigma'$ such that if Σ is a finite branch and $\Sigma' \subset \Sigma$ then there is $i \in S_{\bar{m}}$ such that $V_\Sigma \in \mathscr{T}_i$. (The proof of this claim is analogous to the proof just given; we omit the details.) We choose such $\bar{m}$ and Σ', we choose $k < \omega$ such that $(2/3)^k < 1/\bar{m}$, and we choose a finite branch $\tilde{\Sigma}$ such that

$$\Sigma' \subset \tilde{\Sigma} \text{ and } |\tilde{\Sigma}| \geq \bar{n} + 1.$$

We denote by $\tilde{s}$ the largest element of $\tilde{\Sigma}$ and we denote by $\tilde{n}$ that integer n such that $\tilde{s} \in T_n$.

We recall from the definition of the tree T that for every $t \in T$ there is $B_t \subset \beta$ such that $[B_t]^2$ is the set of immediate successors of t. We set

$$W(0) = \{\tilde{s}\} \quad \text{and}$$
$$W(p) = \bigcup_{t \in W(p-1)} [B_t]^2 \quad \text{for } 1 \leq p \leq k.$$

Now for $0 \leq p \leq k$ we define a family $\{P_{p,i} : i < \beta_{\bar{n}}\}$ such that

$W(p) = \cup_{i<\beta_{\bar n}} P_{p,i}$. (We begin with $p = k$ and we proceed by 'downward induction'.)

For $t \in W(k)$ let $\Sigma(t)$ be the (finite) branch of T determined by t (i.e., $\Sigma(t) = \{s \in T : s \preccurlyeq t\}$) and let

$$t \in P_{k,i} \quad \text{if } V_{\Sigma(t)} \in \mathscr{T}_i.$$

Let $0 < p \leq k$ and assume that the sets $\{P_{p,i} : i < \beta_{\bar n}\}$ have been defined; we define $\{P_{p-1,i} : i < \beta_{\bar n}\}$. Let $t \in W(p-1)$. There is $n < \omega$ such that $t \in T_n$; we have $\bar n < n$. There is $B_t \subset \beta$ with $|B_t| = \beta_{n+1}$ such that the immediate successors of t are the elements of the set

$$[B_t]^2 \subset W(p) = \bigcup_{i<\beta_{\bar n}} P_{p,i}.$$

Since β_{n+1} and $(\beta_{\bar n})^+$ are regular cardinals and $(\beta_{\bar n})^+ \ll \beta_{n+1}$, we have from Theorem 1.5(a) the arrow relation $\beta_{n+1} \to (\beta_{\bar n}^+)^2_{\beta_{\bar n}}$ and *a fortiori* $\beta_{n+1} \to (3)^2_{\beta_{\bar n}}$. Hence there are $C \subset B_t$ with $|C| = 3$ and $i < \beta_{\bar n}$ such that $[C]^2 \subset P_{p,i}$; we let $t \in P_{p-1,i}$.

The definition of $\{P_{p,i} : i < \beta_{\bar n}\}$ for $p \leq k$ is complete.

There is $\bar i < \beta_{\bar n}$ such that $\tilde s \in P_{0,\bar i}$.

It follows from the definition of the families $\{P_{p,i} : p \leq k, i < \beta_{\bar n}\}$ that there is $S \subset T_{\bar n + k}$ with $|S| = 3^k$ such that

$$\tilde\Sigma \subset \Sigma(t) \quad \text{for } t \in S,$$
$$\text{cal}\,\{V_{\Sigma(t)} : t \in S\} \leq 2^k \quad \text{and}$$
$$V_{\Sigma(t)} \in \mathscr{T}_{\bar i} \quad \text{for } t \in S.$$

We have

$$1/\bar m < \kappa(\mathscr{T}_{\bar i}) \leq \text{cal}\,\{V_{\Sigma(t)} : t \in S\}/|S| \leq (2/3)^k < 1/\bar m,$$

a contradiction.

The proof is complete.

6.30 *Corollary.* Let α be an infinite cardinal. There is an extremally disconnected, compact Hausdorff space X such that X has property (*) (hence, X is a c.c.c. space) but for every set $\{\mu_i : i < \alpha\}$ of regular Borel probability measures on X there is a non-empty, open subset U of X such that $\mu_i(U) = 0$ for all $i < \alpha$.

Proof. Since there is a strong limit cardinal $\beta > \alpha$ such that $\text{cf}(\beta) = \omega$, this is immediate from Theorem 6.29.

Remark. For an infinite cardinal β, we say that a space X has *measure-calibre* β if for every family $\{U_\xi : \xi < \beta\}$ of non-empty, open subsets of X there are $B \subset \beta$ with $|B| = \beta$ and a regular, Borel probability measure μ on X such that $\mu(U_\xi) > 0$ for $\xi \in B$. (This notion may be compared with the notion of compact-calibre β described above in Chapter 2 and considered below in Chapter 8.)

It is clear that a space with calibre β, or a space with the property of Lemma 6.28(b), has measure-calibre β. It can be proved (for β a strong limit cardinal with $\mathrm{cf}(\beta) = \omega$) that the space X of Theorem 6.29 does not even have measure-calibre β.

The Galvin–Hajnal example

6.31 *Lemma.* Let $\kappa \geq \omega$. There is a family $\{S_\eta : \eta < 2^\kappa\}$ such that

$$S_\eta \subset \eta \quad \text{for } \eta < 2^\kappa,$$
$$[S_\eta]^2 \subset \bigcup_{\xi < 2^\kappa} (S_\xi \times \{\xi\}) \quad \text{for } \eta < 2^\kappa,$$
$$\text{type } S_\eta \leq \kappa \quad \text{for } \eta < 2^\kappa, \text{ and}$$
$$\text{if } S \subset 2^\kappa \text{ with } [S]^2 \subset \cup_{\xi < 2^\kappa} (S_\xi \times \{\xi\})$$

and type $S \leq \kappa$, then there is $\eta < 2^\kappa$ such that $S = S_\eta$.

Proof. Set $\mathscr{A} = \{A \subset 2^\kappa : \text{type } A \leq \kappa\}$ and let

$$\{A_\eta : \eta < 2^\kappa\}$$

be a well-ordering of $\mathscr{A}$ such that

$$|\{\eta : \eta < 2^\kappa, A = A_\eta\}| = 2^\kappa \quad \text{for } A \in \mathscr{A}.$$

Recursively for $\eta < 2^\kappa$ we define the sets S_η. We set

$$S_0 = \varnothing.$$

If $\eta < 2^\kappa$ and S_ξ has been defined for $\xi < \eta$, we define S_η. We consider two cases.

Case 1. $A_\eta \subset \eta$ and $[A_\eta]^2 \subset \cup_{\xi < \eta} \{\{\zeta, \xi\} : \zeta \in S_\xi\}$. We set

$$S_\eta = A_\eta.$$

Case 2. Case 1 fails. We set

$$S_\eta = \varnothing.$$

The definition of S_η for $\eta < 2^\kappa$ is complete.

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We have $S_\eta = A_\eta$ or $S_\eta = \varnothing$ for $\eta < 2^\kappa$ and hence

$$\text{type } S_\eta \leq \kappa;$$

further for $[S_\eta]^2 \neq \varnothing$ we have

$$[S_\eta]^2 = [A_\eta]^2 \subset \bigcup_{\xi<\eta} \{\{\zeta,\xi\}:\zeta \in S_\xi\} \subset \bigcup_{\xi<2^\kappa} (S_\xi \times \{\xi\}).$$

Finally if $S \subset 2^\kappa$ with type $S \leq \kappa$ and if

$$[S]^2 \subset \bigcup_{\xi<2^\kappa} (S_\xi \times \{\xi\})$$

then (since $\mathrm{cf}(2^\kappa) > \kappa$) there is $\bar{\xi} < 2^\kappa$ such that

$$[S]^2 \subset \{\{\zeta,\xi\}:\zeta \in S_\xi, \xi < \bar{\xi}\}.$$

There is $\bar{\eta}$ such that $\bar{\xi} \leq \bar{\eta} < 2^\kappa$ and $S = A_{\bar{\eta}}$ and from

$$[A_{\bar{\eta}}]^2 = [S]^2 \subset \bigcup_{\xi<\bar{\eta}} \{\{\zeta,\xi\}:\zeta \in S_\xi\}$$

we have

$$S_{\bar{\eta}} = A_{\bar{\eta}} = S,$$

as required.

The proof is complete.

Remark. Let X and Y be spaces and let κ be an infinite cardinal. If $X \leq Y$ and Y satisfies the (κ,κ)-chain condition, then X satisfies the (κ,κ)-chain condition; in particular, X satisfies the (κ,κ)-chain condition if and only if its Gleason space $G(X)$ satisfies the (κ,κ)-chain condition.

(This remark is proved by imitating the proof of Theorem 6.2(c). We omit the details.)

6.32 *Theorem.* Let κ be an infinite cardinal. There is an extremally disconnected, compact Hausdorff space X such that

(i) X has calibre α for every regular cardinal $\alpha > \kappa$; and

(ii) X does not satisfy the (κ,κ)-chain condition.

Proof. It follows from the remark above and Corollary 2.22 that it is enough to find a space X with (i) and (ii); for then the Gleason space of X will have all the required properties.

Let X as a set be equal to $\{0,1\}^{2^\kappa}$, set

$$V_\eta = \{x \in X : x|S_\eta \equiv 0, x_\eta = 1\} \quad \text{for } \eta < 2^\kappa,$$

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and give X the topology determined by the subbase $\{V_\eta : \eta < 2^\kappa\}$.

We verify that the space X satisfies (i) and (ii).

(i) Let α be regular with $\alpha > \kappa$ and let $\{U_\xi : \xi < \alpha\}$ be a family of non-empty, basic open subsets of X; for $\xi < \alpha$ there is a finite subset $F(\xi)$ of 2^κ such that

$$U_\xi = \bigcap_{\eta \in F(\xi)} V_\eta.$$

For $\xi < \alpha$ there is $x \in U_\xi$ and since $x|F(\xi) \equiv 1$ and $x|(\cup_{\eta \in F(\xi)} S_\eta) \equiv 0$ we have

$$(\bigcup_{\eta \in F(\xi)} S_\eta) \cap F(\xi) = \varnothing.$$

For $A \subset \alpha$ we set $F(A) = \cup_{\xi \in A} F(\xi)$. It is enough to show there is $A \subset \alpha$ with $|A| = \alpha$ such that

$$(\bigcup_{\eta \in F(A)} S_\eta) \cap F(A) = \varnothing.$$

We assume without loss of generality, using the Erdős–Rado theorem for quasi-disjoint families (Theorem 1.4), that there is a set J such that

$$F(\xi) \cap F(\xi') = J \quad \text{for } \xi < \xi' < \alpha;$$

we set

$$B(\xi) = F(\xi) \backslash J \quad \text{for } \xi < \alpha$$

and we assume without loss of generality, using the fact that α is a regular, uncountable cardinal, that there is $n < \omega$ such that

$$|B(\xi)| = n \quad \text{for } \xi < \alpha.$$

If $n = 0$ then $F(\xi) = J = F(0)$ for $\xi < \alpha$ and with $A = \alpha$ we have $F(A) = F(0)$ and hence

$$(\bigcup_{\eta \in F(A)} S_\eta) \cap F(A) = (\bigcup_{\eta \in F(0)} S_\eta) \cap F(0) = \varnothing.$$

We assume $n > 0$, we let

$$B(\xi) = \{\eta_\xi^1 < \eta_\xi^2 < \ldots < \eta_\xi^n\}$$

be the listing of the elements of $B(\xi)$ in the natural order inherited from 2^κ, and we choose $B \subset \alpha$ such that

$$|B| = \alpha \quad \text{and}$$
$$\eta_\xi^k < \eta_{\xi'}^k \quad \text{for } 1 \leq k \leq n, \xi < \xi', \xi, \xi' \in B.$$

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We set

$$g(\eta) = \{\zeta \in B : B(\zeta) \cap S_\eta \neq \varnothing\} \quad \text{for } \eta < 2^\kappa, \text{ and}$$
$$f(\xi) = \cup \{g(\eta) : \eta \in B(\xi)\} \quad \text{for } \xi \in B.$$

Since type $S_\eta \leq \kappa$ for $\eta < 2^\kappa$ and

$$B(\zeta) \cap B(\zeta') = \varnothing \quad \text{for } \zeta, \zeta' \in B, \zeta \neq \zeta',$$

we have

$$\text{type } g(\eta) \leq \kappa \quad \text{for } \eta < 2^\kappa$$

and hence

$$\text{type } f(\xi) \leq \kappa \cdot n \text{ (ordinal product) for } \xi \in B.$$

Since $\xi \notin f(\xi)$ for $\xi \in B$, it follows from (the case $\rho = \kappa \cdot n$ of) Theorem 1.11 that there is $A \subset B$ such that

$$|A| = \alpha, \text{ and}$$
$$\zeta \notin f(\xi) \quad \text{for } \xi, \zeta \in A.$$

Now suppose there are $\xi, \zeta \in A$ and $\eta \in F(\xi)$ such that $S_\eta \cap F(\zeta) \neq \varnothing$. Then $\xi \neq \zeta$ and from $\eta \notin F(\zeta)$ we have $n \notin J$ and hence $\eta \in B(\xi)$; further from

$$S_\eta \cap J \subset S_\eta \cap F(\xi) = \varnothing$$

we have $S_\eta \cap B(\zeta) \neq \varnothing$ and hence

$$\zeta \in g(\eta) \subset f(\xi),$$

a contradiction. Thus $(\cup_{\eta \in F(A)} S_\eta) \cap F(A) = \varnothing$, as required.

(ii) It is enough to show that if $\mathscr{T}^*(X) = \cup_{i<\kappa} \mathscr{T}_i$ then there are $\bar{i} < \kappa$, and $A \subset 2^\kappa$ with $|A| = \kappa$, such that

$$\{V_\eta : \eta \in A\} \subset \mathscr{T}_{\bar{i}}, \quad \text{and}$$
$$[A]^2 \subset \bigcup_{\xi < 2^\kappa} (S_\xi \times \{\xi\});$$

indeed then if there is $x \in V_\eta \cap V_{\eta'}$ with $\eta, \eta' \in A$ and $\eta < \eta'$ we have from $\{\eta, \eta'\} \in S_{\eta'} \times \{\eta'\}$ that $\eta \in S_{\eta'}$ and hence $x_\eta = 0$ (since $x \in V_{\eta'}$) and $x_\eta = 1$ (since $x \in V_\eta$), a contradiction.

We fix $f : \kappa \to \kappa$ such that

$$|f^{-1}(\{i\})| = \kappa \quad \text{for } i < \kappa$$

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and we define a subset $\{\eta_\xi : \xi < \kappa\}$ of 2^κ such that

$\eta_0 = 0$;

$\{\eta_{\xi'} : \xi' < \xi\} \subset S_{\eta_\xi}$ for $\xi < \kappa$; and

if $\xi < \kappa$ and $f(\xi) = i$, and if there is $\eta < 2^\kappa$ such that $V_\eta \in \mathscr{T}_i$ and $\{\eta_{\xi'} : \xi' < \xi\} \subset S_\eta$, then $V_{\eta_\xi} \in \mathscr{T}_i$.

We proceed by recursion.

We set $\eta_0 = 0$.

If $\xi < \kappa$ and $\eta_{\xi'}$ has been defined for $\xi' < \xi$, then

$$[\{\eta_{\xi'} : \xi' < \xi\}]^2 \subset \bigcup_{\eta < 2^\kappa} (S_\eta \times \{\eta\})$$

and hence there is $\eta < 2^\kappa$ such that $\{\eta_{\xi'} : \xi' < \xi\} \subset S_\eta$. We choose for η_ξ such an ordinal η; if there is such an ordinal η so that (in addition) $V_\eta \in \mathscr{T}_{f(\xi)}$, we choose η_ξ to satisfying also this condition.

The definition of the set $\{\eta_\xi : \xi < \kappa\}$ is complete. Since

$$\eta_{\xi'} < \eta_\xi \quad \text{for } \xi' < \xi < \kappa$$

we have type $\{\eta_\xi : \xi < \kappa\} = \kappa$ and hence there is $\bar{\eta} < \kappa$ such that

$$\{\eta_\xi : \xi < \kappa\} = S_{\bar{\eta}}.$$

There is $\bar{i} < \kappa$ such that $V_{\bar{\eta}} \in \mathscr{T}_{\bar{i}}$. We set

$$A = \{\eta_\xi : \xi < \kappa, f(\xi) = \bar{i}\}$$

and we note that

$$|A| = \kappa,$$
$$V_\eta \in \mathscr{T}_{\bar{i}} \quad \text{for } \eta \in A, \text{ and}$$
$$[A]^2 \subset [S_{\bar{\eta}}]^2 \subset \bigcup_{\eta < 2^\kappa} (S_\eta \times \{\eta\}),$$

as required.

The proof is complete.

6.33 *Corollary*. There is an extremally disconnected, compact Hausdorff space X such that

(i) every uncountable, regular cardinal number is a calibre for X, and

(ii) X does not have a strictly positive measure (and in fact X does not have property (*)).

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Proof. This follows from Lemmas 6.2 and 6.22(a) and (the case $\kappa = \omega$ of) Theorem 6.31.

6.34 *Problem.* Is there a compact Hausdorff space satisfying the (ω, ω)-chain condition but not condition $(*)$?

Property $(**)$ for non-compact spaces

Every space with a strictly positive measure has property $(**)$, and the converse holds for compact, Hausdorff spaces (cf. Theorem 6.4). We show now that this converse fails for appropriate completely regular, Hausdorff spaces; the examples we define have precalibre α for every cardinal α such that $\mathrm{cf}(\alpha) > \omega$.

The first of these examples is σ-compact (and hence normal); the second, the Pixley–Roy space of 6.37, is not normal and is first-countable.

6.35 *Theorem.* There is a completely regular, Hausdorff space X such that

(a) X does not have a strictly positive measure;

(b) X has property $(**)$ (hence $G(X)$ and βX have strictly positive measures); and

(c) X has precalibre α for every cardinal α with $\mathrm{cf}(\alpha) > \omega$.

Proof. Using the notation of Chapter 4, we set

$$X = \Sigma_\omega \{0, 1\}^{\omega^+},$$

i.e.,

$$X = \{x \in \{0, 1\}^{\omega^+} : |\{\eta < \omega^+ : x_\eta \neq 0\}| < \omega\}.$$

We verify statements (a), (b) and (c).

(a) We suppose there is a strictly positive measure μ on X, we choose a pseudobase $\mathscr{B}$ for X such that $\mu(B) > 0$ for all $B \in \mathscr{B}$, we set

$$U_\xi = \{x \in X : x_\xi = 1\} \quad \text{for } \xi < \omega^+,$$

and for $\xi < \omega^+$ we choose $B_\xi \in \mathscr{B}$ such that $B_\xi \subset U_\xi$. There are $\varepsilon > 0$ and $A \subset \omega^+$ with $|A| = \omega^+$ such that

$$\mu(B_\xi) \geq \varepsilon \quad \text{for } \xi \in A.$$

Let $\{\xi(n): n < \omega\}$ be a subset of A faithfully indexed by ω and set

$$S_k = \bigcap_{n \geq k} B_{\xi(n)} \quad \text{for } k < \omega.$$

From the definition of the set X it follows that $\cap_{k<\omega} S_k = \varnothing$, and from Theorem C.1 of the Appendix we have $\mu(\cap_{k<\omega} S_k) \geq \varepsilon$. This contradiction completes the proof of (a).

(b) The space $\{0, 1\}^{\omega^+}$, a product of separable spaces, is a Hausdorff compactification of X. The space $\{0, 1\}^{\omega^+}$ has a strictly positive measure by Corollary 6.8, and the statements of (b) follow from Theorem 6.4 and Corollary 6.6.

(c) If $\mathrm{cf}(\alpha) > \omega$ then $\{0, 1\}^{\omega^+}$ has calibre α (and hence precalibre α) by Theorem 3.18(a). Hence the space X, a dense subspace of $\{0, 1\}^{\omega^+}$, has precalibre α (cf. Theorems 2.16(b) and 2.17(b)).

The proof is complete.

We note that it follows from the Hewitt–Marczewski–Pondiczery theorem (B.2 of the Appendix) that the space $\{0, 1\}^{\omega^+}$ is itself separable. That $\{0, 1\}^{\omega^+}$ has property (**) is then immediate, and statement (c) of Theorem 6.35 follows from Theorem 2.7.

That the space X of Theorem 6.35 is σ-compact is a special case of Lemma 8.9 below. For a direct argument one may set

$$X_n = \{x \in \{0, 1\}^{\omega^+} : |\{\eta < \omega^+ : x_\eta \neq 0\}| \leq n\} \quad \text{for } n < \omega$$

and note that

X_n is closed in $\{0, 1\}^{\omega^+}$ (hence X_n is compact), and
$X = \bigcup_{n<\omega} X_n$.

We define the Pixley–Roy space. For a non-empty, finite subset F of $\mathbb{R}$ and an open subset V of $\mathbb{R}$, we denote by $U(F, V)$ the set

$$U(F, V) = \{G : F \subset G \subset V, |G| < \omega\}.$$

We note that if F is not a subset of V then $U(F, V) = \varnothing$.

Definition. The elements of the Pixley–Roy space X are the non-empty, finite subsets of $\mathbb{R}$. The topology $\mathscr{T}(X)$ is the topology generated by the subbase

$$\mathscr{S}(X) = \{U(F, V) : F \in X, V \in \mathscr{T}^*(\mathbb{R})\}.$$

6.36 *Lemma.* The Pixley–Roy space X has these properties:
(a) the subbase $\mathscr{S}(X)$ is a base for (the topology of) X;
(b) the elements of $\mathscr{S}(X)$ are open-and-closed in X;
(c) X is a completely regular, Hausdorff space.

Proof. (a) It is enough to show that if $\{U(F_k, V_k): k < n\} \subset \mathscr{S}(X)$ with $n < \omega$ then $\cap_{k<n} U(F_k, V_k) \in \mathscr{S}(X)$. Set

$$F = \bigcup_{k<n} F_k \text{ and } V = \bigcap_{k<n} V_k.$$

Then $\cap_{k<n} U(F_k, V_k) = U(F, V) \in \mathscr{S}(X)$, as required.

(b) We show that if $U(F, V) \in \mathscr{S}(X)$, then $U(F, V)$ is closed in X. If $G \in X \backslash U(F, V)$, then one of the inclusions $F \subset G$, $G \subset V$ must fail. If $F \not\subset G$ there is $p \in F \backslash G$ and then $U(G, \mathbb{R} \backslash \{p\})$ is a neighborhood of G such that

$$U(F, V) \cap U(G, \mathbb{R} \backslash \{p\}) = \varnothing,$$

and if $G \not\subset V$ then $U(G, \mathbb{R})$ is a neighborhood of G such that

$$U(F, V) \cap U(G, \mathbb{R}) = \varnothing.$$

(c) It is clear that for $F \in X$ we have

$$\{F\} = \cap \{U(F, V): F \subset V \in \mathscr{T}(\mathbb{R})\},$$

and hence $\{F\}$ is a closed subset of X. Statement (c) now follows from (a) and (b).

6.37 *Theorem.* Let X be the Pixley–Roy space. Then
(a) X does not have a strictly positive measure;
(b) βX is separable (hence βX has a strictly positive measure and X has property $(**)$); and
(c) X has precalibre α for every cardinal α with $\mathrm{cf}(\alpha) > \omega$.

Proof. (a) We suppose there is a strictly positive measure μ on X, we choose a pseudobase $\mathscr{B}$ for X such that $\mu(B) > 0$ for all $B \in \mathscr{B}$, and for $F \in X$ we choose $B(F) \in \mathscr{B}$ such that

$$B(F) \subset U(F, \mathbb{R}).$$

We set

$$X_n = \{F \in X: \mu(B(F)) \geq 1/n\} \text{ for } n < \omega$$

and we note that since $X = \cup_{n<\omega} X_n$ there is $\bar{n} < \omega$ such that $|X_{\bar{n}}| \geq \omega$. Let $\{F_n: n < \omega\}$ be a subset of $X_{\bar{n}}$ faithfully indexed by ω,

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and set

$$S_k = \bigcup_{n \geq k} B(F_n) \text{ for } k < \omega, \text{ and}$$

$$S = \bigcap_{k<\omega} S_k.$$

From Theorem C.1 of the Appendix we have $\mu(S) \geq 1/\bar{n}$ and hence there is $F \in S$. For $k < \omega$ there is $n(k) \geq k$ such that

$$F \in B(F_{n(k)}) \subset U(F_{n(k)}, \mathbb{R})$$

and hence

$$\bigcup_{k<\omega} F_{n(k)} \subset F;$$

it follows that $|F| = \omega$, a contradiction.

(b) Let $\mathscr{V} = \{V_n : n < \omega\}$ be a countable base for the usual topology of $\mathbb{R}$ such that $\mathscr{V}$ is closed under finite unions, and for $n < \omega$ set

$$\mathscr{U}_n = \{U(F, V) : F \in X, V \in \mathscr{T}(\mathbb{R}), F \subset V_n \subset V\}.$$

It is then clear, retaining the notation

$$\mathscr{S}(X) = \{U(F, V) : F \in X, V \in \mathscr{T}^*(\mathbb{R})\},$$

that $\mathscr{U}_n \subset \mathscr{S}(X)$ for $n < \omega$. Further, since $\mathscr{V}$ is a basis for $\mathbb{R}$ closed under finite unions, for $F \in X$ and $V \in \mathscr{T}^*(\mathbb{R})$ with $F \subset V$ there is $n < \omega$ such that

$$F \subset V_n \subset V$$

and hence $U(F, V) \in \mathscr{U}_n$; it follows that

$$\mathscr{S}(X) = \bigcup_{n<\omega} \mathscr{U}_n.$$

For $n < \omega$ the set $\mathscr{U}_n$ has the finite intersection property; indeed if $m < \omega$ and

$$U(F(k), V(k)) \in \mathscr{U}_n \text{ for } k < m,$$

then with

$$F = \bigcup_{k<m} F(k), \qquad V = \bigcap_{k<m} V(k)$$

we have

$$\bigcap_{k<m} U(F(k), V(k)) = U(F, V) \in \mathscr{U}_n.$$

Hence for $n < \omega$ there is

$$p_n \in \cap \{\mathrm{cl}_{\beta X} U : U \in \mathscr{U}_n\}.$$

We claim that $\{p_n : n < \omega\}$ is dense in βX.

If W is a non-empty, open subset of βX there is a non-empty open subset of βX such that

$$S \subset \mathrm{cl}_{\beta X} S \subset W,$$

there is $U(F, V) \in \mathscr{S}(X)$ such that

$$\varnothing \neq U(F, V) \subset S \cap X,$$

and there is $n < \omega$ such that $U(F, V) \in \mathscr{U}_n$; it follows that

$$p_n \in \mathrm{cl}_{\beta X} U(F, V) \subset \mathrm{cl}_{\beta X}(S \cap X) \subset W,$$

as required.

The proof that βX is separable is complete. The remaining statements of (b) follow from 6.2–6.7 above.

(c) For $\mathrm{cf}(\alpha) > \omega$, the (separable) space βX has calibre α (by Theorem 2.7); hence X has precalibre α.

The proof is complete.

We note in passing that the Pixley–Roy space is first-countable. Indeed for $F \in X$ set

$$V_n = \cup \{(x - 1/n, x + 1/n) : x \in F\};$$

then $\{U(F, V_n) : n < \omega\}$ is a countable local neighborhood basis in X at F.

Another class of examples of spaces with property (**), but with no strictly positive measure, described by Koumoullis, is discussed in the Notes to this chapter.

6.38 *Problem.* Find a topological characterization of the (completely regular, Hausdorff) spaces with a strictly positive measure.

A possible characterization is the following condition, a strengthening of (**) suggested by the examples of 6.35 and 6.37: The set $\mathscr{T}^*(X)$ can be written in the form $\mathscr{T}^*(X) = \cup_{n<\omega} \mathscr{T}_n$ where, for $n < \omega, \kappa(\mathscr{T}_n) > 0$ and

$$\text{if } \{U_k : k < \omega\} \subset \mathscr{T}_n \text{ then } \cap_{m<\omega} [\cup_{k>m} U_k] \neq \varnothing.$$

6.39 *Problem* (Negrepontis). Is there a compact c.c.c. space X,

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with no strictly positive measure, that is *point-homogeneous* (in the sense that for $p, q \in X$ there is a homeomorphism h of X onto X such that $h(p) = q$)?

Recall in this connection that every compact topological group has a (strictly positive) Haar probability measure.

Notes for Chapter 6

The study of strictly positive measures originated in the theory of Boolean algebras (especially in the work of Tarski) and in topological measure theory; for older references the reader should consult the text of Sikorski (1964), especially section 42. Our definition of a (space with a) strictly positive measure, which appears at first glance unorthodox, was dictated by two considerations: (a) The definition should be sufficiently general to ensure Theorem 6.7 and (b) for compact Hausdorff spaces the definition should be equivalent to the property that there exist a regular Borel probability measure assigning positive measure to every non-empty open set.

Lemmas 6.2(a), (b) and 6.3, and Theorem 6.4 ((b)$\Leftrightarrow$(c)) for the case of totally disconnected compact Hausdorff spaces, are due to Kelley (1959). The equivalence ((a)$\Leftrightarrow$(d)) of Theorem 6.4 has also been noted by Hebert & Lacey (1968, Theorems 4.6 and 4.12).

The implication (a)$\Rightarrow$(d) of Theorem 6.4, that if a compact space X has a strictly positive measure then its Gleason space $G(X)$ does also, can be proved without any appeal to condition (**) or the relation $\equiv$. The canonical projection from $G(X)$ onto X furnishes an embedding of $C(X)$ into $C(G(X))$, and the given (strictly positive) measure on X furnishes a positive linear functional Λ on $C(X)$ such that $\Lambda(1) = 1$. There is by the Hahn–Banach theorem a positive linear extension $\bar{\Lambda}$ of Λ defined on $C(G(X))$, and by the Riesz representation theorem (C.10 of the Appendix) there is a regular Borel probability measure $\bar{\mu}$ on $G(X)$ such that

$$\bar{\Lambda}(f) = \int_{G(X)} f\,d\bar{\mu} \quad \text{for } f \in C(G(X));$$

that $\bar{\mu}$ is strictly positive on $G(X)$ follows from the fact that the set $\mathscr{R}(X)$ of regular-open subsets of X is identified, via the Stone representation of $\mathscr{R}(X)$, with a base for $G(X)$. A similar argument, using the fact that every element of $C^*(X)$ extends to an element of $C^*(\beta X)$ (Theorem B.5(a) of the Appendix), shows that if a completely regular, Hausdorff space has a strictly positive measure then its Stone–Čech compactification does also.

For $0<\theta\leq 1$, a space X is said to have *property* $(**)_\theta$ if the set $\mathscr{T}^*(X)$ can be written in the form $\mathscr{T}^*(X)=\cup_{n<\omega}\mathscr{T}_n$ with $\kappa(\mathscr{T}_n)\geq\theta$ for $n<\omega$.

For a compact, Hausdorff space X we denote by $M(X)$ the set of finite, regular Borel measures on X with the w^*-topology (as dual of the Banach space $C(X)$), and by $M_1^+(X)$ and $S_{M(X)}$ the subset of probability measures and the unit ball, respectively. The following statements have been proved.

(a) X is separable if and only if X has property $(**)_1$.

(b) (Fakhoury 1976, Mägerl & Namioka 1980). The following conditions are equivalent.

(1) X has property $(**)_\theta$ for some $\theta, 0<\theta<1$;

(2) X has property $(**)_\theta$ for all $\theta, 0<\theta<1$;

(3) $M_1^+(X)$ is w^*-separable;

(4) $S_{M(X)}$ is w^*-separable;

(5) there is an isomorphic embedding of $C(X)$ into $l^\infty=C(\beta\mathbb{N})$; and

(6) there is an isometric embedding of $C(X)$ into l^∞.

(c) $M(X)$ is separable if and only if there is a one-to-one continuous, linear operator from $C(X)$ into l^∞.

(d) X is separable $\Rightarrow X$ has property $(**)_\theta$ for some (or all) θ, $0<\theta<1\Rightarrow M(X)$ is w^*-separable $\Rightarrow X$ has property $(**)$.

(e) The space Ω of Theorem 6.9 (and of Theorem 6.18 for $\beta=\omega$), which is the space Ω for which $C(\Omega)=L[0, 1]$, is a non-separable, extremally disconnected Hausdorff space with property $(**)_\theta$ for $0<\theta<1$ such that, assuming $\omega^+=2^\omega$, Ω does not have calibre ω^+.

(f) Talagrand (1980b) has shown, assuming $\omega^+=2^\omega$, that there is a compact, Hausdorff space Ω such that $M(\Omega)$ is w^*-separable and $M_1^+(\Omega)$ is not w^*-separable (i.e., Ω does not have property $(**)_\theta$ for $0<\theta<1$).

(g) Kalamidas has shown that if $\{X_i : i\in I\}$ is a set of compact, Hausdorff spaces with property $(**)_\theta$ for $0<\theta<1$ and if $|I|\leq 2^\omega$, then the product space $\prod_{i\in I}X_i$ also has property $(**)_\theta$.

(h) There are of course spaces Ω with property (**) such that $M(\Omega)$ is not w^*-separable. (Indeed if $M(\Omega)$ is w^*-separable then, as is obvious from (c) above, we have $w(M(\Omega)) \leq 2^\omega$; but for every cardinal α there is Ω with property (**) such that $w(\Omega) \geq \alpha$.) It follows from Pełczynski (1958) and Rosenthal (1970b) (Theorem 4.8 and following Remarks) that with Ω the space such that $C(\Omega) = L^\infty(\{0, 1\}^{\omega^+})$, the space Ω has property (**) and weight equal to 2^ω and $M(\Omega)$ is not w^*-separable.

(i) (Argyros & Negrepontis 1982).

(1) X has a strictly positive measure if and only if $M_1^+(X)$ has a strictly positive measure;

(2) if $\alpha > \omega^+$ with α regular and X has a strictly positive measure, then $M_1^+(X)$ has calibre α.

(3) assuming $\omega^+ = 2^\omega$, the space X defined by Haydon (1978) (for another purpose) has a strictly positive measure and $M_1^+(X)$ does not have calibre ω^+.

(4) for $2 \leq n < \omega$, X has property K_n if and only if $M_1^+(X)$ has property K_n.

(5) X is productively c.c.c. if and only if $M_1^+(X)$ is productively c.c.c.

(6) X^I is c.c.c. for all sets I if and only if $M_1^+(X)$ is c.c.c.

(7) $M_1^+(X)$ has calibre ω^+ if and only if for every family $\{V_\xi : \xi < \omega^+\}$ of non-empty, open subsets of X, and for $0 < \theta < 1$, there is $A \subset \omega^+$ with $|A| = \omega^+$ such that $\kappa(\{V_\xi : \xi \in A\}) \geq \theta$.

We cite two related problems. Here as above, X remains a compact, Hausdorff space.

Problem (Mägerl & Namioka, 1980). If $M(X)$ is w^*-separable, must $M(G(X))$ be w^*-separable? (Here as usual $G(X)$ denotes the Gleason space of X.)

Problem (Rosenthal, 1970a). If X is extremally disconnected and $M(X)$ is w^*-separable, must $M_1^+(X)$ be w^*-separable?

The measure-theoretic argument of Corollary 6.8 was introduced by Oxtoby (1961, statement 2.7) to show that the product of separable spaces has the countable chain condition (cf. also the case $\kappa = \omega, \alpha = \omega^+$ of Theorem 3.28 above).

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The material in 6.12–6.17 is due to Argyros & Kalamidas (1981); for regular $\alpha > \omega$, Corollary 6.17 has been known for some years to Kunen. Lemma 6.12(i) is, of course, an immediate consequence of Jensen's elementary inequality (see for example Spivak (1967, p. 199)).

It was noted by Erdős many years ago, but not published at the time, that assuming the continuum hypothesis the Stone space Ω of the measure algebra of Lebesgue measure on $[0, 1]$ does not have calibre ω^+; the more general statement in Theorem 6.18 is due to Argyros & Tsarpalias (1981). Four proofs of Erdős' theorem are given by Tall (1974) and Kunen & Tall (1979); in the former paper it is noted (Example 7.5) that if Martin's axiom is assumed then Ω does not have calibre 2^ω.

Assuming $\omega^+ = 2^\omega$, Haydon (1978) has given a construction in functional analysis, roughly related to Erdős' example, which answers in the negative a conjecture of Pełczynski (1968). This example has proved useful on various occasions; the (projective limit) construction that yields this example apparently has considerable similarity with the following constructions (all assuming CH): the result of Kunen (1981) establishing the existence of a compact L-*space* (i.e. a non-separable, hereditarily Lindelöf space); an example of Losert (1979, Theorem 3) on the existence of a compact separable space with a probability measure μ such that μ has a uniformly distributed sequence, but μ does not have a well-distributed sequence; and the example of Talagrand (1980a) mentioned above, establishing the existence of a compact space X such that $M(X)$ is w^*-separable but $M^1_+(X)$ is not w^*-separable. Fleissner & Negrepontis (1979) have found a model of $\mathrm{ZFC} + (\omega^+ < 2^\omega)$ in which Haydon's construction can be repeated. This raises the question, probably to be answered positively, whether it is possible to find, in some situation where the continuum hypothesis fails, a compact Hausdorff space with a strictly positive measure and without calibre ω^+.

The following statement, in its essentials a special case of Theorem 6.21, is attributed by Knaster (1945, footnote 11) to Marczewski ($\equiv$ Szpilrajn) (1945): Of any ω^+ subsets of $\mathbb{R}$ with positive Lebesgue measure, some ω^+ have pairwise non-empty intersection. See in this connection the later paper of Marczewski (1947, 1.1 (iii)).

That every Stone space $\Omega = S(\mathscr{A})$ for $\mathscr{A}$ a measure algebra has property (K) (a special case of Lemma 6.22) is due to Horn & Tarski (1948) (Theorem 2.4 (i)$\Rightarrow$(iii)); it is noted by Halmos (1963, Section 24, Exercise 2) that Ω is not separable if $(X, \mathscr{S}, \lambda)$ is atomless.

Theorem 6.22(b) is due to Kalamidas.

Rosenthal (1970a) has studied the existence of a strictly positive measure on a (compact, Hausdorff) space X in relation to the Banach space $C(X)$ of real-valued, continuous functions on X and its dual space $M(X)$ of regular Borel measures on X. He has shown for example that

(a) if X is a compact, Hausdorff space with c.c.c. and $C(X)$ is a conjugate Banach space, then X has a strictly positive measure (Rosenthal 1970a, Theorem 4.1); and

(b) a compact, Hausdorff space X has a strictly positive measure if and only if there is a weakly compact subspace of $M(X)$ whose convex hull is weak*-dense in $M(X)$ (Rosenthal 1970a, Theorem 4.5(b)).

The example of 6.23, due to Gaifman (1964), was for a long time the only space known with the countable chain condition and with no strictly positive measure. Parts (c) and (d) of Theorem 6.23 are due to Kalamidas and Negrepontis, respectively. A simplified version of Gaifman's example is given by Fremlin (1980).

Theorems 6.25, 6.26 and 6.29 are due to Argyros (1981d).

Rubin & Shelah (1981) have also constructed, assuming the continuum hypothesis, for $2 \le m < \omega$ a (compact, Hausdorff, extremally disconnected) space which has property K_m and does not have property K_{m+1}. Their construction, motivated by certain model-theoretic considerations, is related both to property (c) of Lemma 7.7 (and the resulting property (i) of Theorem 7.9) and to property (ii) of Lemma 7.12 (and the resulting property $S(X) \le \alpha^+$ in Theorem 7.13).

The constructions of Gaifman and Argyros can be generalized as follows.

Replacing in Theorem 6.23 the space $\mathbb{R}$ by the Stone space of the α-homogeneous, α-universal Boolean algebra, whose essential properties are given by Negrepontis (1969), Negrepontis (1981) proved that for $\alpha = \alpha^{\underline{\alpha}}$ there is a (generalized Gaifman) extremally disconnected, compact Hausdorff space X such that

(a) X has property $(*(\alpha))$ – i.e., the set $\mathscr{T}^*(X)$ can be written in the form $\mathscr{T}^*(X) = \cup_{i<\alpha} \mathscr{T}_i$ where, for $i < \alpha$, no k_i elements of the family $\mathscr{T}_i$ are pairwise disjoint (here $\{k_i : i < \alpha\}$ is a set of natural numbers);

(b) X does not have property $(**(\alpha))$ – i.e., the set $\mathscr{T}^*(X)$ cannot be written in the form $\mathscr{T}^*(X) = \cup_{i<\alpha} \mathscr{T}_i$ with $\kappa(\mathscr{T}_i) > 0$ for $i < \alpha$;

(c) X has property $K_{\alpha^+,n}$ for $2 \leq n < \omega$; and

(d) assuming $\alpha^+ = 2^\alpha$, X does not have calibre α^+.

Replacing in Theorem 6.26 the set $\{0, 1\}^\omega$ by the space $(\{0, 1\}^\alpha)_\alpha$, whose essential properties are given by Hung & Negrepontis (1973, 1974), Argyros and Negrepontis proved that for $\alpha = \alpha^{\underline{\alpha}}$ and $2 \leq m < \omega$ there is a (generalized Argyros) extremally disconnected, compact Hausdorff space X such that

(a) X has property $(*(\alpha))$;

(b) X does not have property $(**(\alpha))$;

(c) X has property $K_{\alpha^+,m}$; and

(d) assuming $\alpha^+ = 2^\alpha$, X does not have property $K_{\alpha^+,m+1}$.

For $2 \leq m < \omega$, a space X is *σ-m-linked* if the set $\mathscr{T}^*(X)$ of non-empty, open subsets of X can be written in the form $\mathscr{T}^*(X) = \cup_{n<\omega} \mathscr{T}_n$ where, for $n < \omega$, every m elements of $\mathscr{T}_n$ have non-empty intersection. It has been shown by Bell (1982b), in ZFC with no special set-theoretic assumptions, that for $2 \leq m < \omega$ there is a compactification $B\omega$ of ω such that the remainder $B\omega \backslash \omega$ is *σ-m*-linked but not σ-$(m + 1)$-linked.

The work of Gaifman (1964) left unresolved the question, posed by Horn & Tarski (1948, p. 482), whether every c.c.c. space has property $(*)$. The example of 6.32, due to Galvin & Hajnal (1981), settles a strong (and generalized) form of this question; in addition to the properties given in the statement of Theorem 6.32, we note that the space X defined there can be proved to have density character κ^+ and to have both weight and pseudoweight equal to 2^κ. We note further that, using the free-set theorem of Hajnal (1961) for singular cardinals (a modification of Theorem 1.11 not proved in this monograph) and also the extension to singular cardinals of the Erdős–Rado theorem on quasi-disjoint families (Theorem 1.9), the Galvin–Hajnal space of Theorem 6.32 has, for example in the case $\kappa = \omega$, calibre α whenever $\mathrm{cf}(\alpha) > \omega$.

A preliminary and weak version of Theorem 6.32, based in part on the work of Erdős, Galvin & Hajnal (1975), was obtained some years ago by Hajnal (unpublished notes).

We thank Fred Galvin for providing us with historical information and notes concerning Theorem 6.32.

That the σ-compact space $\sum_{\omega}\{0, 1\}^{\omega^+}$ has the properties given in Theorem 6.35 was proved by Argyros (October, 1979).

The Pixley–Roy space of Lemma 6.36 was introduced by Pixley & Roy (1969) in connection with the study of Moore spaces. Our proofs of Lemma 6.36 and Theorem 6.37(b) follow closely work of van Douwen (1977a, b), who has defined and investigated several classes of spaces of Pixley–Roy type and computed a number of cardinal invariants associated with them (including the density character of the associated Stone–Čech compactification); independently A. Sapounakis observed (August, 1979) that the Pixley–Roy space does not have a strictly positive measure and its Stone–Čech compactification does.

G. Koumoullis has observed (November, 1979) that every non-separable Banach space X, with its weak topology (induced by its dual space X^*), satisfies property (∗∗) but does not have a strictly positive measure.

Indeed, X with this topology satisfies property (∗∗) because it is homeomorphic to a dense subspace of $\mathbb{R}^{(X^*)}$ (cf. 2.16(b) and (6.2(d)). If X has a strictly positive measure μ, then the unit ball S of X^* with the weak* topology satisfies the following conditions: S is homeomorphic to a subspace of $\mathbb{R}^X$ (in the product topology); S is compact (and convex); and if $f, g \in S$ and $f = g$ μ-almost everywhere, then $f = g$. According to a theorem of A. I. Tulcea (1974), it now follows that S with the weak* topology is metrizable. Since X is a subspace of $C(S)$ with the uniform norm and $C(S)$ is separable, it follows that X itself is separable, a contradiction.

The following remarks point to a connection between the spaces with a strictly positive measure and the geometric theory of Banach spaces.

A norm $\| \quad \|$ on a linear space B is said to be *strictly convex* if $x, y \in B$ with $x \neq y$ and $\|x\| = \|y\| = 1$ imply $\|(x+y)/2\| < 1$; a Banach space is *strictly convexifiable* if it has an equivalent norm that is strictly convex.

It is a difficult problem in the geometry of Banach spaces to determine those compact spaces X for which the Banach space $C(X)$ is strictly convexifiable (cf. Dashiell & Lindenstrauss, 1973). We outline now a proof that $C(X)$ is strictly convexifiable whenever X has a strictly positive measure. First, every space of the form $L^\infty(\mu)$, for μ a finite measure, is the dual of $L^1(\mu)$, a weakly compactly generated space; by the theorem of Amir & Lindenstrauss (1968), there are a set Γ and a one-to-one bounded linear operator $T: L^\infty(\mu) \to c_0(\Gamma)$. It then follows from Day's theorem (1955) and a remark of Klee (1953) that $L^\infty(\mu)$ is strictly convexifiable. Now if X has a strictly positive measure μ, then $C(X)$ is a subspace of $L^\infty(\mu)$ in a natural way and is therefore itself strictly convexifiable. (A more direct proof, also noted by R. R. Phelps, is as follows. Let μ be a strictly positive measure on X, identify $C(X)$ in the natural way with a subspace of $L^2(X, \mu)$, let $\| \ \|_u$ and $\| \ \|_2$ denote respectively the uniform and the L^2 norms on $C(X)$, and define $\|f\| = \|f\|_u + \|f\|_2$ for $f \in C(X)$. Then $\| \ \|$ is a compatible norm for $C(X)$, and $\| \ \|$ is strictly convex because $\| \ \|_2$ is strictly convex. A simple extension of this argument shows that a compact space has a strictly positive measure if and only if there is a one-to-one positive continuous linear function from $C(X)$ into a space of the form $L^2(Y, \nu)$ with ν a finite measure on Y.)

It can be verified for the spaces X of Argyros given in Theorem 6.26 that $C(X)$ is not strictly convexifiable; Argyros' spaces are to our knowledge the only examples known of c.c.c. spaces X for which $C(X)$ is not strictly convexifiable. These remarks suggest this question, posed by Argyros and Negrepontis: Does every compact c.c.c. space X such that $C(X)$ has an equivalent strictly convex norm have a strictly positive measure? We indicate in the Notes for Chapter 7, using results on Corson-compact spaces, that this question, assuming the continuum hypothesis, has a negative answer.